








Modelling Financial Markets

Chapter 3: Factor Models and Asset Pricing



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Why Do Assets Have Different Average Returns?

What we will explain

- What explains expected returns?
  - the Capital Asset Pricing Model (CAPM) and multifactor models
- How are exposures estimated?
  - betas, their uncertainty and shrinkage
  - shrinkage: pulling a noisy estimate towards a target value
- How are pricing restrictions tested?
  - a joint test of the alphas, Fama–MacBeth regressions, model comparison
- Do the findings survive? Multiple testing and replication
- Which factors can we find directly in the data? PCA and IPCA

What you should be able to do

- estimate a beta with heteroskedasticity- and autocorrelation-robust errors and shrink it
- test the CAPM alphas jointly and estimate risk premia with robust errors
- judge a new factor against multiple-testing hurdles and replication evidence
- extract latent factors by PCA and choose how many to keep
- decompose a factor ETF into factor exposures and alpha

Route through the Chapter

1. What Explains Expected Returns?
2. From Markowitz to the CAPM
3. Estimating Beta
4. Testing the CAPM
5. Multifactor Models
6. Fama–MacBeth Regressions
7. Discount Factors and Model Comparison
8. The Factor Zoo and Multiple Testing
9. Latent Factors: PCA and IPCA
10. Factor Investing
11. Case Study: Is There a Replication Crisis?
12. AI (Artificial Intelligence) for Scientific Discovery
13. Conclusions

- Answers to the four quick checks are in the Appendix
  - the checks of Sections 3–4 and those of Sections 6–8

What Is a Factor Model?

- A **factor model** explains the return of asset i by a few common variables $f_{1,t}, \ldots, f_{K,t}$, the **factors**

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1} f_{1,t} + \cdots + \beta_{i,K} f_{K,t} + \varepsilon_{i,t}$$

- $R_{i,t}$: return of asset i in period t
- $R_{f,t}$: risk-free rate
- K: number of factors
- α_i: average return not explained by the factors, 0 if the factors price the asset correctly
- $\beta_{i,k}$: **factor loading** (exposure) of asset i to factor k
- $\varepsilon_{i,t}$: asset-specific (idiosyncratic) shock

What Is a Factor Model Used For?

- Two uses in this chapter
  - Explain an asset: the technology ETF (exchange-traded fund) XLK moves 1.28 times the US market (Section 3)
  - Evaluate a fund: is its α_i positive once its exposures are priced (Sections 4 and 10)?
- Other uses: cost of capital, portfolio construction and risk management (Campbell, Lo & MacKinlay (1997), chapters 5–6)
- The risk-management advantage: far fewer parameters
  - N assets have $N(N+1)/2$ variances and covariances
  - a K-factor model needs only the loadings, the factor covariance and N specific variances
  - in total: $NK + K(K+1)/2 + N$ parameters
- Key message: expected returns should be explained by **exposures to priced risks**, not by the name of the asset

The Data: Which Series We Use

Series	What it contains	Frequency	Period
Fama–French factors	returns on five factors: market, size, value, profitability and investment	monthly	1963–2026
Momentum factor	stocks with the highest returns over months $t-12$ to $t-2$ minus those with the lowest	monthly	1963–2026
25 portfolios	US stocks sorted independently into 5 size groups and 5 B/M (book-to-market) groups; the 25 intersections form the portfolios	monthly	1963–2026
Sector ETFs	one ETF for each sector of the S&P 500 (index of 500 large US companies): 9 sectors from 1999, 11 from 2018	daily	1999–2026
Factor ETFs	momentum, value, quality, minimum volatility (from 2013); Russell size, value and growth (from 2000)	daily	2000–2026
Bucharest Stock Exchange (BVB)	11 large stocks (blue chips) and the exchange index, BET	daily	2015–2026

- Sector ETF tickers
  - XLB materials, XLE energy, XLF financials, XLI industrials, XLK technology
  - XLP consumer staples, XLU utilities, XLV health care, XLY consumer discretionary
  - from 2018 also XLRE real estate and XLC communication services

The Data: Sources and Notation

Sources

- Factors and the 25 portfolios: Kenneth French Data Library, monthly returns in percent
- Last month available: July 2026 (for momentum: August 2026)
- ETFs and stocks: daily adjusted closing prices (dividends reinvested)
- Last day available: 18 September 2026

Notation

- **BET**: the BVB benchmark index, a price index (no dividends)
- **Excess return**: $R^e_{i,t} = R_{i,t} - R_{f,t}$
- $R_{f,t}$: one-month US Treasury bill rate (RF in the French data)
- **Mkt-RF**: excess return of the US stock market, weighted by market capitalisation

- Why these data
  - the very different mean returns of the 25 portfolios make the test demanding
  - the ETFs are investable; the BVB applies the methods to the local market

Which Returns This Chapter Uses

- Notation: simple and log returns
  - $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$ (simple return) and $r_{i,t} = \ln(1 + R_{i,t})$ (log return)
  - $P_{i,t}$: the price adjusted for dividends and splits
- Regressions on the French factors use monthly simple returns
  - The factors are portfolio returns, and a portfolio return is computed from simple returns
  - The mean log return would be lower by about $\sigma^2/2$: at an annual volatility of 20%, about 2 percentage points a year
- Rolling betas, BVB betas and principal component analysis (Section 9) use daily log returns
- At the BVB the daily risk-free rate is treated as a constant, so beta is the same with raw or excess returns

Which Portfolio Does a Mean–Variance Investor Hold?

- The question of this section: if investors choose only by mean and variance, which portfolio do they hold?
- The second question: what mean return does each asset then earn?
- Step 1: the Markowitz (1952) frontier gives the portfolio with the highest Sharpe ratio
- Step 2: the CAPM (Sharpe (1964); Lintner (1965)) says that this portfolio is the market portfolio itself
- We apply both to the 9 US sector ETFs: they are investable and together cover the S&P 500

The Mean–Variance Frontier

- Markowitz (1952): with weights w, expected excess returns $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, the frontier solves

$$\min_{\mathrm{w}} \ \mathrm{w}^{\top}\boldsymbol{\Sigma}\,\mathrm{w} \quad \text{subject to} \quad \mathrm{w}^{\top}\boldsymbol{\mu} = \mu^{*},\ \mathrm{w}^{\top}1 = 1$$

- Negative weights are **short positions**
- The solution: $\mathrm{w}^{*} = \boldsymbol{\Sigma}^{-1}(\eta_1\boldsymbol{\mu} + \eta_2 1)$
  - η_1, η_2: Lagrange multipliers set by the two constraints
- The weights w^{*} are linear in μ^{*}
  - every frontier portfolio is a combination of two frontier portfolios (two-fund separation)
- In the (σ_p, μ_p) plane the frontier is a hyperbola
  - the branch above the minimum-variance portfolio is the **efficient frontier**

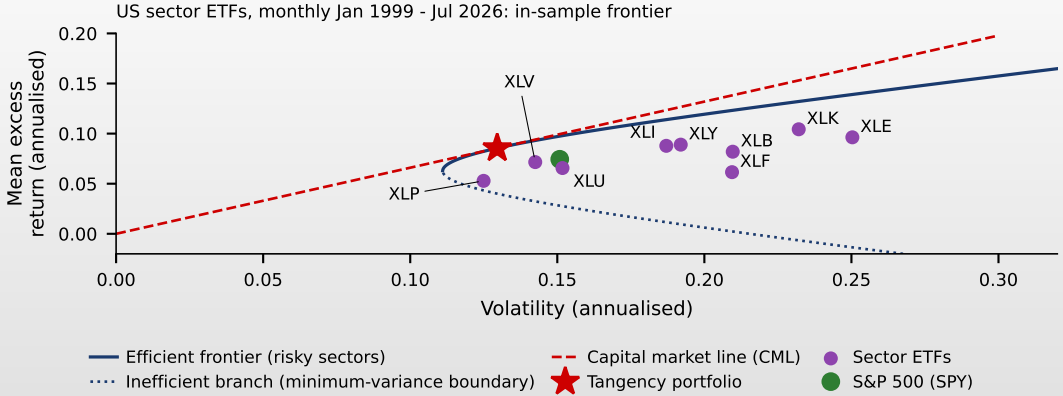
The Sharpe Ratio and the Tangency Portfolio

- **Sharpe ratio** $SR_p = \mathbb{E}[R_p - R_f]/\sigma_p$: the excess return per unit of volatility
- **Tangency portfolio**: the portfolio with the highest Sharpe ratio

$$w_{\text{tan}} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} \,/\, 1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$$

  - the formula holds when $\boldsymbol{\Sigma}$ is positive definite and the denominator is positive
  - R_p, σ_p: return and volatility of portfolio p
  - the denominator rescales $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$ so that the weights sum to 1
- **CML** (Capital Market Line): all mixes of the risk-free asset and the tangency portfolio, $\mu_p = SR_{\text{tan}}\,\sigma_p$
- If the market is the tangency portfolio, every asset's mean return is linear in its beta
  - Seminar 3, problem A2

The Efficient Frontier of US Sectors (1/2)



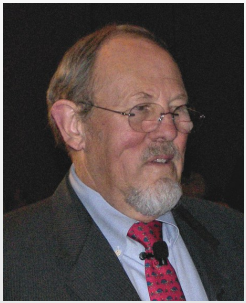
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The Efficient Frontier of US Sectors (2/2)

- Chart axes: annualised volatility and mean excess return, as fractions ($0.10 = 10\%$)
  - 9 sector ETFs, monthly, January 1999 – July 2026 ($T = 331$ months)
  - the frontier is estimated in-sample
- Tangency portfolio (red star): mean 8.55%, volatility 12.96%, Sharpe ratio 0.66
- SPY (green circle): mean 7.44%, volatility 15.08%, Sharpe ratio 0.49
  - SPY: the ETF that tracks the S&P 500, used as the market proxy
- The CAPM (next slides) requires the market to be the tangency portfolio itself
  - in this sample SPY lies below the frontier
- The gap is not significant
  - the GRS test of Section 4, with Mkt-RF instead of SPY as the market, gives $p = 0.85$
- The tangency portfolio holds large short positions
  - -0.37 in financials and -0.21 in materials (and $+0.37$ in health care)
- The extreme weights arise because the means are estimated with noise (Chapter 4)

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The CAPM: Sharpe, Lintner, Black



William F. Sharpe, Nobel Prize in Economics 1990, with Markowitz and Miller

Photo: Larry D. Moore (2007); CC BY 4.0; Wikimedia Commons

- The **CAPM** was formulated by Sharpe (1964) and Lintner (1965)
- All investors hold the same tangency portfolio
  - if they all choose mean–variance efficient portfolios and share the same beliefs
- Since every asset must be held by someone, the tangency portfolio is the **market portfolio**
- Expected excess return is proportional to **beta**

$$\mathbb{E}[R_i] - R_f = \beta_i \left(\mathbb{E}[R_m] - R_f\right), \qquad \beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}$$

- R_i, R_m: returns of asset i and of the market portfolio
- $\mathbb{E}[R_m] - R_f$: the **market risk premium**
- β_i: sensitivity of asset i to the market
  - $\beta_i > 1$ amplifies market moves
- Black (1972): a version without unlimited borrowing at the risk-free rate
  - the zero-beta CAPM (Section 4)

What the CAPM Implies

- Assumptions
  - mean–variance investors with identical beliefs, one period
  - unlimited borrowing and lending at R_f
  - no taxes or trading costs
- Only beta is priced: no other characteristic should predict returns
- Idiosyncratic risk earns nothing, because diversification removes it
- **SML** (Security Market Line): every asset lies on the line (beta, mean)
  - its slope equals the market premium
  - it is the testable statement of the CAPM (Section 4)
- Unlike the SML, the CML contains only efficient portfolios
- Which portfolio is "the market"?
  - the answer decides what a test of the CAPM can establish (Roll's critique, Section 4)

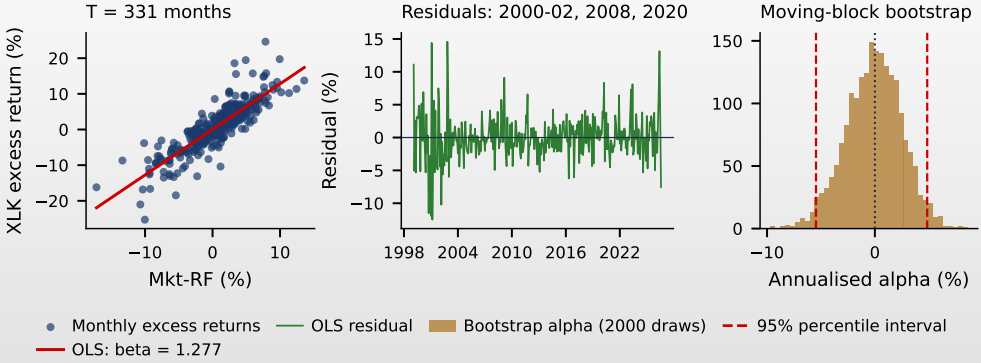
The Cost of Capital: Which Uncertainty Matters?

- CAPM cost of capital: $R_f + \hat{\beta}_i \, (\widehat{\mathbb{E}[R_m]} - R_f)$
- The standard error (SE) of the market premium: $15.4\%/\sqrt{63} = 1.9\%$ a year
  - 15.4%: annual market volatility
  - 63: years of data
- Multiplied by the XLK beta, this gives an uncertainty of 2.5% a year in the cost of capital
  - XLK beta: $\hat{\beta} = 1.28$ (Section 3)
- The uncertainty of beta adds only $0.082 \times 7.2\% = 0.6\%$ a year
  - the SE of beta times the premium
- Conclusion: the uncertainty of the market premium weighs far more than that of beta

How Precisely Can We Measure Beta?

- The question of this section
  - by how much does an asset move when the market moves by 1%?
  - how sure is that number?
- Beta enters the cost of capital and hedging
- The running example: XLK, the technology ETF, the sector with the highest beta in the sample
- The steps
  - robust standard errors and the choices that change beta
  - rolling betas, shrinkage, the BVB application

The Beta of the Technology ETF (XLK) in the Data



Left: the OLS line on the monthly returns; middle: the OLS residuals, with spikes in 2000–2002, 2008 and 2020; right: the bootstrap distribution of alpha, with its 95% interval

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The Market Model and HAC Errors

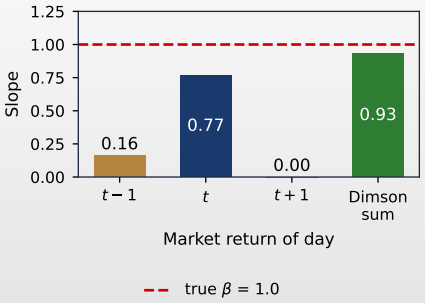
- Market model $R^e_{i,t} = \alpha_i + \beta_i R^e_{m,t} + \varepsilon_{i,t}$
- **OLS** (Ordinary Least Squares): $\hat{\beta}_i = \widehat{\mathrm{Cov}}(R^e_i, R^e_m)/\widehat{\mathrm{Var}}(R^e_m)$
- Variance split $\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma^2_{\varepsilon,i}$
  - systematic risk (priced under the CAPM) plus idiosyncratic risk (diversifiable)
  - σ_i^2, σ_m^2: variances of the asset and of the market; $\sigma^2_{\varepsilon,i}$: residual variance
  - $R^2 = \beta_i^2 \sigma_m^2 / \sigma_i^2$ is the systematic share
- Why **HAC** (heteroskedasticity and autocorrelation consistent) errors (Newey & West (1987))
  - volatility clustering (Chapter 1) makes the residuals heteroskedastic
  - the product $R^e_{m,t}\varepsilon_{i,t}$ can be autocorrelated
  - both make the classical standard error of $\hat{\beta}$ too small
  - Bartlett weights and $L = \lfloor 4(T/100)^{2/9} \rfloor$ lags (Newey & West (1994))
  - $T = 331$ gives $L = 5$; $T = 757$ gives $L = 6$
  - the point estimates remain those of OLS

Choices That Change an Estimated Beta (1/2)

- Frequency and window: usually 3–5 years of monthly data or 1–2 years of daily data
  - a longer window lowers the standard error but mixes different beta regimes (Betas Are Not Constant)
- Market proxy: SPY (S&P 500) or Mkt-RF for US assets, BET for Romanian stocks
- Data cleaning: an unadjusted capital event creates a fake return
  - Example: 7 September 2023 is the ex-date of a RON 1.7225 per share special dividend of Fondul Proprietatea (FP)
  - On that day the close falls by 81%, but the total return of the day is +6.0%

Choices That Change an Estimated Beta (2/2)

- Asynchronous trading: the closing prices of different assets are not recorded at the same moment
- An illiquid stock reacts to market news with a delay, so the daily OLS beta is biased towards 0
- The Dimson beta (Dimson (1979)): regression on three days of market returns
  - the market return of the previous day, the same day and the next day
  - $R_{i,t} = a + b_{-1} R_{m,t-1} + b_0 R_{m,t} + b_{+1} R_{m,t+1} + \varepsilon_{i,t}$
  - the slopes are added: $\beta^D = b_{-1} + b_0 + b_{+1}$
- The chart shows a simulation
  - a stock with true beta 1 trades on only 75% of days
- The slope on the same-day market return is 0.77, so it understates beta
- The slope on the previous-day market return is 0.16: part of the reaction arrives one day late
- The Dimson beta is 0.93, much closer to 1



Worked Example: Beta of the Technology ETF (1/2)

		Standard error		
	Estimate	**Classical**	**White**	**Newey–West**
α (annual)	0.33%	2.27%	2.32%	2.45%
β	1.277	0.041	0.054	0.082

- XLK on Mkt-RF, monthly January 1999 – July 2026, $T = 331$, $R^2 = 0.742$
- Columns of the table
  - **Classical**: errors assumed homoskedastic and serially uncorrelated
  - **White**: robust to heteroskedasticity (White (1980))
  - **Newey–West**: also robust to autocorrelation, $L = 5$ lags
- The robust SE of β is twice the classical one
- Heteroskedasticity explains part of the increase (White column)
  - autocorrelation of the product $R^e_{m,t}\varepsilon_t$ explains the rest

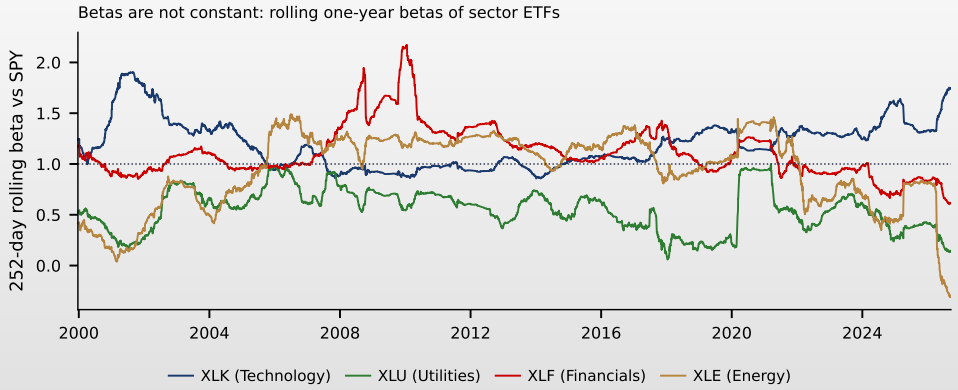
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Worked Example: Beta of the Technology ETF (2/2)

- t-statistic of alpha: $t(\alpha) = 0.33/2.45 = 0.13$
- 95% bootstrap interval for alpha: $[-5.5\%, 4.9\%]$
  - bootstrap SE 2.61%, slightly above 2.45%
  - moving-block bootstrap of 12-month blocks of the pairs $(R^e_{\text{XLK},t}, R^e_{m,t})$, which keeps the dependence over time
  - 2 000 resamples (Seminar 3, problem B1)

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Betas Are Not Constant (1/2)



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Betas Are Not Constant (2/2)

- The chart: rolling OLS beta on 252 daily log returns (about one year)
- The beta of financials (XLF) ranged from 0.60 to 2.17
- The XLF beta peaked right after the 2008–2009 banking crisis
- The beta of technology (XLK) ranged from 0.86 to 1.90
- The XLK beta was high in the dot-com bubble (2000–2001)
  - it is 1.75 at the end of the sample
- The beta of energy (XLE) ranged from -0.31 to 1.49
- The XLE beta is negative at the end of the sample, because oil moved against the market
- The beta of utilities (XLU), the classic defensive sector, ranged from 0.06 to 1.00
- A single full-sample beta hides these regimes
- Models with time-varying betas are covered in Chapters 5–6

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Shrinking Betas: Blume and Vasicek (1/2)

- **Regression to the mean**: a very high or very low estimated beta is partly noise
  - so in the next period it tends to move towards 1
- Blume (1971): regression of betas in one period on betas in the previous period
- Hence the practitioner rule: $\beta^{\text{Blume}} = 0.33 + 0.67\,\hat{\beta}$
- Vasicek (1973): a weighted average of the asset's estimate and a prior

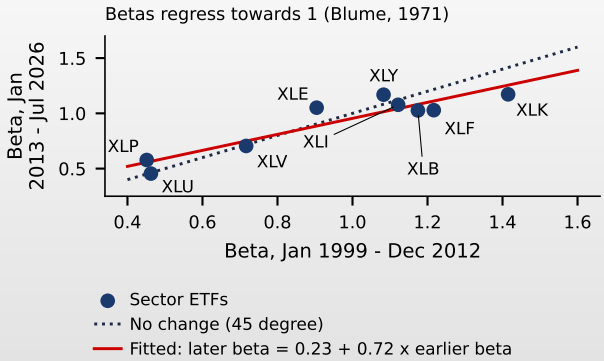
$$\beta_i^V = w_i \hat{\beta}_i + (1 - w_i)\bar{\beta}, \qquad w_i = \frac{\sigma_\beta^2}{\sigma_\beta^2 + \text{se}(\hat{\beta}_i)^2}$$

  - the prior: the mean $\bar{\beta}$ of the betas of all assets
  - w_i: the weight of the estimate, larger when $\hat{\beta}_i$ is precise (small SE)
  - $\text{se}(\hat{\beta}_i)$: its standard error
  - σ_β^2: the variance of the **true** betas across assets

Shrinking Betas: Blume and Vasicek (2/2)

- The variance of the estimates across assets adds true variance and noise
  - $\text{Var}_{cs}(\hat{\beta}) = \sigma^2_\beta + \overline{\text{se}^2}$, with $\overline{\text{se}^2}$ the mean squared standard error
- So $\sigma^2_\beta = \text{Var}_{cs}(\hat{\beta}) - \overline{\text{se}^2}$, the empirical-Bayes estimate
  - derivation: Seminar 3, problem A3
- Example: XLK on SPY, monthly 1999–2012: $\hat{\beta} = 1.415$, Newey–West SE 0.139
  - the benchmark is SPY, hence a different beta from 1.28
- Prior from the 9 sectors: $\bar{\beta} = 0.949$ and $\sigma^2_\beta = 0.1159 - 0.0077 = 0.1082$
- The weight on the own estimate: $w = 0.1082/(0.1082 + 0.139^2) = 0.849$
  - it gives the Vasicek beta $\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$
- Blume beta: $0.33 + 0.67 \times 1.415 = 1.278$
- The beta realised on 2013–2026 was 1.172
  - both adjustments are closer to it than the unadjusted beta, and Blume is the closest

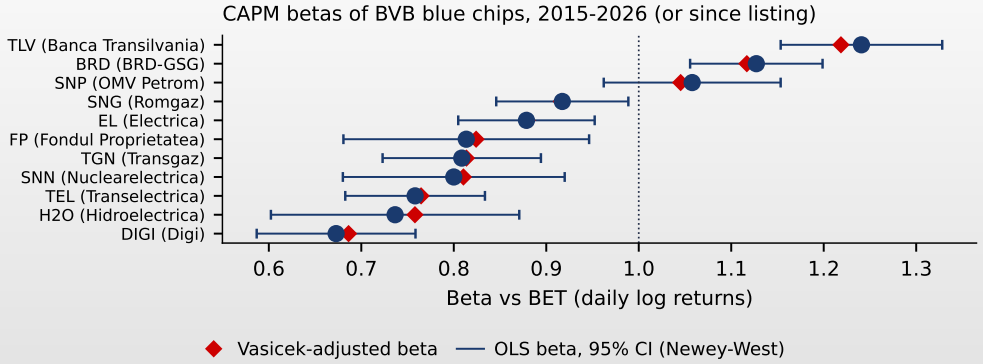
Do Betas Regress Towards One?



- Monthly betas of 9 sector ETFs on SPY: 1999–2012 (horizontal) vs 2013–2026 (vertical)
- Fitted line $\hat{\beta}_{\text{later}} = 0.23 + 0.72\,\hat{\beta}_{\text{earlier}}$
  - very close to Blume's $0.33 + 0.67\,\hat{\beta}$
- Forecast error: the **RMSE** (Root Mean Squared Error)
  - the unadjusted beta has RMSE 0.135
  - the Blume beta has RMSE 0.109
  - the Vasicek beta has RMSE 0.114
- Shrinkage cuts the forecast error by 15–20%
- With only 9 sectors the lesson is qualitative

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CAPM Betas on the Bucharest Stock Exchange (1/3)



Legend: OLS beta with its 95% confidence interval (CI), Newey–West errors; Vasicek-adjusted beta

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CAPM Betas on the Bucharest Stock Exchange (2/3)

- Daily log returns on BET, from 6 January 2015 or the listing date
  - listing dates: DIGI May 2017, H2O July 2023
  - FP prices are adjusted for the special dividend
- Banks are the most market-sensitive
  - Banca Transilvania (TLV) 1.24, with $R^2 = 0.62$
  - BRD 1.13
- The least sensitive
  - Digi 0.67, with $R^2 = 0.21$
  - Hidroelectrica 0.74 and Transelectrica 0.76
- R^2 is high for TLV partly because TLV has a large weight in BET
  - the stock is part of its own benchmark

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CAPM Betas on the Bucharest Stock Exchange (3/3)

- Vasicek shrinkage changes the estimates little
  - daily betas are precise (SE between 0.04 and 0.07)
- Stale prices bias daily betas towards zero
  - the Dimson beta rises from 0.80 to 0.94 for Nuclearelectrica
  - and from 0.81 to 0.905 for Transgaz (Seminar 3, problem B4)
- BET is a price index while the stock prices include dividends
  - alpha is pushed up by roughly beta times the dividend yield of the index
  - beta changes little with BET-TR

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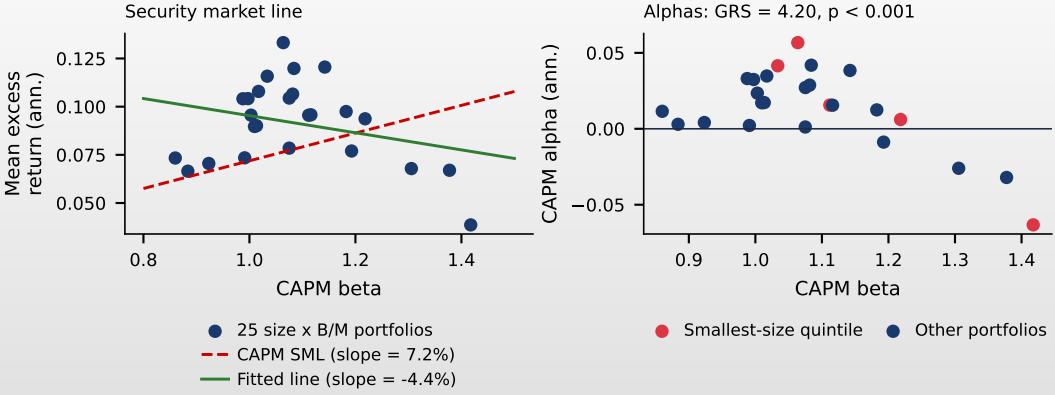
Quick Check: A Beta, Its Error and Its Shrinkage

- A stock, 120 monthly returns: $\hat{\beta} = 0.90$, classical SE 0.05, Newey–West SE 0.09
  1. Give both 95% intervals. Is the stock less risky than the market under each?
  2. Which interval do you report, and why does the point estimate not change?
- A cross section of betas has $\mathrm{Var}_{cs}(\hat{\beta}) = 0.10$, $\overline{\mathrm{se}^2} = 0.02$ and mean $\bar{\beta} = 1.0$
  3. Compute σ^2_{β} and the Vasicek beta of an asset with $\hat{\beta} = 1.6$, once with SE 0.2 and once with SE 0.1
  4. Which of the two is shrunk more, and why? Compare both with the Blume beta
- An illiquid BVB stock, daily returns
- Slopes on BET: 0.25 for the previous day, 0.55 for the same day, 0.00 for the next day
  5. Which Dimson beta do you report?
  6. Why is the same-day slope smaller than the Dimson beta?
- Answers: ▸ Appendix

Is the CAPM Rejected by the Data?

- The question of this section
  - do mean returns lie on the line $\mathbb{E}[R_i^e] = \beta_i \, \mathbb{E}[R_m^e]$?
  - that is, do all assets have zero alpha?
- Test assets: the 25 size $\times$ B/M portfolios (Fama & French (1993))
- Why these portfolios: their mean returns differ widely, so the test has power
  - from 3.9% to 13.3% a year
- The steps
  - the SML in the data
  - the GRS test, which checks all alphas at once
  - the limits of the test

The Security Market Line in the Data (1/2)



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The Security Market Line in the Data (2/2)

- The 25 portfolios, July 1963 – July 2026, $T = 757$
  - re-sorted each June and value-weighted
- The CAPM line has slope 7.2% a year
  - chart axes: annual fractions, $0.05 = 5\%$
- The fitted line has slope -4.4%: higher beta, **lower** average return
  - fitted line: the cross-sectional regression of mean returns on beta
- Betas span only 0.86–1.42
- Alpha is $+5.7\%$ a year for small firms with high B/M (small-value)
  - and -6.3% a year for small firms with low B/M (small-growth)
- 10 of the 25 alphas have $|t| > 1.96$
- The joint test of the 25 alphas (GRS, next slides) gives 4.20
  - $p = 8 \times 10^{-11}$: the CAPM is rejected

MFM_ch3_sml

From One Alpha to the GRS Test (1/2)

- **Jensen's alpha** (Jensen (1968)): the market-model intercept α_i
  - The CAPM implies $\alpha_i = 0$ for every asset
- A separate t-test for each asset: XLK has $t = 0.13$
  - With many assets, some alphas look significant by chance alone (Section 8)
- **The GRS test** (Gibbons, Ross & Shanken (1989)): one test for all assets, $H_0 : \alpha_1 = \cdots = \alpha_N = 0$

$$W = \frac{T - N - 1}{N} \, \frac{\hat{\boldsymbol{\alpha}}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\alpha}}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \;\sim\; F(N,\, T - N - 1)$$

- Notation
  - N: number of test assets; T: number of months; $\hat{\boldsymbol{\alpha}}$: the N estimated alphas
  - $\hat{\boldsymbol{\Sigma}}$: maximum-likelihood residual covariance (divided by T)
  - $\hat{\mu}_m / \hat{\sigma}_m$: Sharpe ratio of the market

From One Alpha to the GRS Test (2/2)

- $\hat{\boldsymbol{\alpha}}^{\top}\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}}$: a weighted sum of squared alphas
  - an alpha counts more when the asset's residuals are less volatile
- Decision: reject H_0 when W exceeds the 95% quantile of $F(N, T-N-1)$
- The distribution is exact, conditional on the factors
  - errors i.i.d. multivariate Normal, independent of the factors
  - zero mean and constant nonsingular covariance
- Interpretation: a large W means the market portfolio is far from the efficient frontier
  - the frontier built from the market and the test assets
- For $N = 1$, $W = t^2$: GRS generalises the t-test
  - t: the OLS t-statistic of alpha
  - $F(1, T-2)$ is the distribution of the square of a Student-t variable with $T-2$ degrees of freedom

A Flat SML: Black and Betting Against Beta (1/2)



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia (2009 upload); public domain; Wikimedia Commons

- Black (1972): without unlimited borrowing at R_f, expected return stays linear in beta
  - but the line starts at the return of a **zero-beta** portfolio, above R_f
- Prediction: the SML is **flatter** than the CAPM line
- Frazzini & Pedersen (2014): investors who want more risk but cannot borrow buy high-beta assets
  - these become expensive and earn lower future returns

A Flat SML: Black and Betting Against Beta (2/2)

- Their strategy, **BAB** (Betting Against Beta):

$$r_t^{\text{BAB}} = \frac{R_{L,t} - R_{f,t}}{\beta_L} - \frac{R_{H,t} - R_{f,t}}{\beta_H}$$

  - $R_{L,t}$, $R_{H,t}$: returns of the low-beta (L) and high-beta (H) portfolios
  - β_L, β_H: their betas
- The strategy is long $1/\beta_L > 1$ units of L and short $1/\beta_H < 1$ units of H
  - the long leg is financed at R_f; the proceeds of the short leg are lent at R_f
  - it has beta zero and a positive average return in their data
- A plain low-minus-high-beta spread, without scaling, has a negative beta
  - its mean mixes the BAB premium with a negative market exposure (Seminar 3, problem C2)

GRS as a Sharpe-Ratio Test

- An algebraic identity (Gibbons, Ross & Shanken (1989)), with all moments estimated by maximum likelihood

$$\hat{\boldsymbol{\alpha}}^{\top}\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}} = \widehat{SR}^2(f,\mathrm{R}) - \widehat{SR}^2(f)$$

  - $\widehat{SR}^2(f,\mathrm{R})$: squared Sharpe ratio of the tangency portfolio of the factor and the N test assets
  - $\widehat{SR}^2(f)$: that of the factor alone
- So GRS measures how much the Sharpe ratio rises when the test assets are added

$$W = \frac{T-N-1}{N}\cdot\frac{\widehat{SR}^2(f,\mathrm{R}) - \widehat{SR}^2(f)}{1+\widehat{SR}^2(f)}$$

- The monthly Sharpe ratio of the market (Mkt-RF) alone is 0.134 (0.47 a year)
- The monthly Sharpe ratio of the tangency portfolio of the market and the 25 portfolios is 0.405 (1.40 a year)
- $W = 29.24 \times (0.405^2 - 0.134^2)/(1+0.134^2) = 4.20$, with $29.24 = (757-25-1)/25$
- Under the CAPM, noise alone would raise the monthly $\widehat{SR}^2$
  - on average by $N(1+\widehat{SR}^2(f))/(T-N-3) = 0.035$
  - observed: 0.146 (▸ Appendix)

The Generalized Method of Moments (1/2) *(deeper)*

- An economic model implies $\mathbb{E}[g_t(\boldsymbol{\theta})] = 0$ for a vector of moment conditions g_t
- **GMM** (Generalized Method of Moments, Hansen (1982)) chooses $\hat{\boldsymbol{\theta}}$ so that the sample mean of the moments is zero

$$\bar{g}(\hat{\boldsymbol{\theta}}) = \frac{1}{T}\sum_{t=1}^{T} g_t(\hat{\boldsymbol{\theta}}) = 0$$

  - exactly identified case: as many moments as parameters
  - with more moments than parameters, not all can be zero, and $\bar{g}$ is made as small as possible with a weighting matrix
- Sandwich variance in the exactly identified case

$$\mathrm{Var}(\hat{\boldsymbol{\theta}}) \approx \frac{1}{T}\, D^{-1} S\, D^{-1\top}, \qquad D = \mathbb{E}\left[\partial g_t / \partial \boldsymbol{\theta}^{\top}\right], \qquad S = \sum_{j=-\infty}^{\infty} \mathrm{Cov}(g_t, g_{t-j})$$

  - D: Jacobian of the moments (their derivatives with respect to the parameters)
  - S: long-run variance of the moments, autocovariances included, estimated by Newey–West (Section 3)

The Generalized Method of Moments (2/2) *(deeper)*

- The simplest case, a mean: $g_t = R_t - \mu$ gives $D = -1$ and $\mathrm{Var}(\hat{\mu}) \approx S/T$
  - this is the HAC t-statistic of a mean factor return (Section 5)
- OLS: $g_t = x_t(y_t - x_t^\top \boldsymbol{\theta})$ gives $D = -\mathbb{E}[x_t x_t^\top]$
  - White errors when S has no lags, Newey–West errors otherwise (Section 3)
- **GRS as GMM**: parameters (α_i, β_i) for $i = 1, \dots, N$
  - $2N$ moments $\varepsilon_t \otimes (1, f_t)$: each residual times 1 and times the factor ($\otimes$: Kronecker product)
  - the system is exactly identified, so the estimates are the OLS ones, asset by asset; GMM changes only their variance
  - ordering the parameters asset by asset, $(\alpha_1, \beta_1, \alpha_2, \beta_2, \dots)$, gives $D = -I_N \otimes (Z^\top Z/T)$, with Z the $T \times 2$ matrix of rows $(1, f_t)$
- $\hat{V}_\alpha$: the block of the sandwich matrix that belongs to the alphas, computed with a HAC $\hat{S}$
- Wald statistic $W_{\mathrm{GMM}} = \hat{\boldsymbol{\alpha}}^\top \hat{V}_\alpha^{-1} \hat{\boldsymbol{\alpha}} \to \chi^2_N$ under H_0
- The same machinery gives the misspecification-robust premia errors (Appendix)
  - it also estimates the discount-factor model of Section 7

When the GRS Assumptions Fail

- **Heavy tails and volatility clustering** (Chapter 1): the F distribution is no longer exact
  - Residual bootstrap under H_0: resample whole rows of residuals, rebuild returns with $\alpha = 0$ (Seminar 3, problem B7)
  - The bootstrap is robust to non-Normality only if the errors are i.i.d. and independent of the factors
  - GMM Wald test (Section 4), with a HAC $\hat{V}_\alpha$: robust to heteroskedasticity and autocorrelation
- **Too many assets for the number of months**: $\hat{\boldsymbol{\Sigma}}$ has rank at most $T - 2$
  - when $N \geq T - 1$, $\hat{\boldsymbol{\Sigma}}$ cannot be inverted and GRS cannot be computed
  - the power of the test falls well before that limit
- With many assets there are tests that do not invert $\hat{\boldsymbol{\Sigma}}$ (Pesaran & Yamagata (2024); ▸ Appendix)

Roll's Critique

- Roll (1977): the CAPM is a statement about the **true** market portfolio of all wealth
- The stock indices used as the market omit bonds, real estate, private firms and human capital
- Mathematically, betas computed against **any** efficient portfolio price all mean returns exactly
  - Seminar 3, problem A2
- So a CAPM test really checks whether the index used as the market is efficient
- A rejection can come from the model or from the choice of index

The Choice of Test Assets Changes the Conclusion

- On the 9 sector ETFs of Section 2 the CAPM is not rejected
  - January 1999 – July 2026, $T = 331$, with Mkt-RF as the market
  - GRS $= 0.53$ with $p = 0.85$
- Adding the sectors raises the squared monthly Sharpe ratio by 0.015
  - less than the 0.029 that noise alone would produce (Appendix)
- Over the same 331 months the 25 size $\times$ B/M portfolios reject it
  - GRS $= 2.91$, $p = 9 \times 10^{-6}$
- The reason: the sectors differ little in the characteristics that explain returns
  - each sector mixes small and big, value and growth firms
- Conclusion: failing to reject a model is weak evidence in its favour
  - test models on assets whose mean returns differ widely

Quick Check: Reading a GRS Statistic

- One factor, $N = 4$ test assets, $T = 65$ months
- Monthly Sharpe ratios: factor alone 0.15, tangency portfolio of the factor and the 4 assets 0.25
  1. Compute W from the Sharpe-ratio form. The 95% quantile of $F(4, 60)$ is about 2.53: reject?
  2. Compute the squared Sharpe-ratio gain expected under H_0, $N(1 + \widehat{SR}^2(f))/(T - N - 3)$, and compare it with the observed gain
  3. Same Sharpe ratios, $T = 650$ months: recompute W (95% quantile of $F(4, 645)$: about 2.39). What changed, the gap or the evidence?
- Answers: Appendix

Arbitrage Pricing Theory

- The market beta does not explain mean returns (Section 4)
  - the question of this section: how many factors do we need?
  - and which ones?
- **APT**: Arbitrage Pricing Theory, Ross (1976)
- Assumption: returns follow a K-factor model

$$R_{i,t} = \mathbb{E}[R_i] + \boldsymbol{\beta}_i^\top \mathsf{f}_t + \varepsilon_{i,t}, \qquad \mathbb{E}[\mathsf{f}_t] = 0$$

  - asset-specific risks can be diversified away
  - $\boldsymbol{\beta}_i$: the K loadings of asset i
  - f_t: the unexpected part of the factors
- **No arbitrage** (no sure profit without capital invested) implies $\mathbb{E}[R_i] - R_f = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$
- The relation is approximate: it holds for almost all assets, not exactly for each one
- $\boldsymbol{\lambda}$: the **risk premia**, the extra expected return per unit of each exposure
- The theory does not say **which** factors: they must be found in the data

Building Factors: Long–Short Portfolios (1/2)

size: NYSE median; B/M: 30% / 40% / 30%

	Low (growth)	Neutral	High (value)
Small	S_L	S_N	S_H
Big	B_L	B_N	B_H

$$\text{SMB} = \tfrac{1}{3}(S_L + S_N + S_H) - \tfrac{1}{3}(B_L + B_N + B_H)$$

$$\text{HML} = \tfrac{1}{2}(S_H + B_H) - \tfrac{1}{2}(S_L + B_L)$$

- A **characteristic-sorted factor** buys stocks with a high value of a characteristic
  - and sells those with a low value
- The short side finances the long side
  - so the factor return is a difference of returns, $f_t = R_{\mathbf{high},t} - R_{\mathbf{low},t}$
- Fama–French: each June, stocks are split into 2×3 groups
  - using the previous year's accounting data
  - each group is a value-weighted portfolio
- The size breakpoint is the median market capitalisation of NYSE (New York Stock Exchange) stocks

Building Factors: Long–Short Portfolios (2/2)

- NYSE breakpoints and value weights stop microcaps from dominating the portfolios
  - **SMB** (Small Minus Big): small minus big stocks
  - **HML** (High Minus Low): high B/M (value) minus low B/M (growth), averaged over small and big firms, hence at the same size
- The same construction on profitability and investment gives RMW and CMA (next slide)

The Standard Factor Models

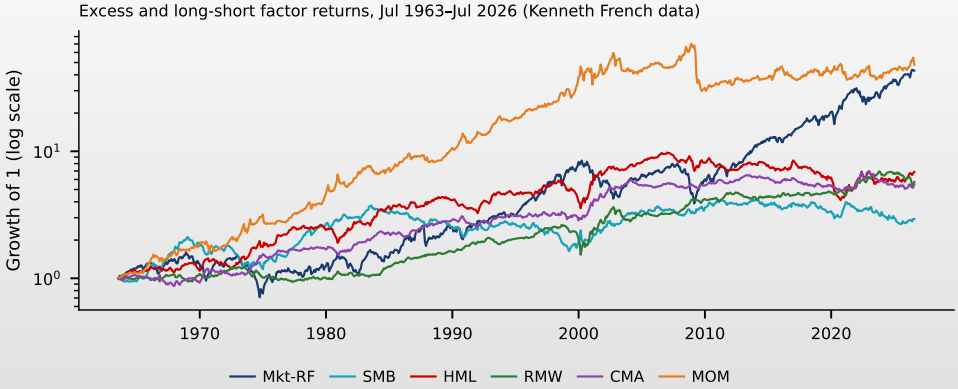
Model	Factors	Reference	Motivation
CAPM	Mkt-RF	Sharpe (1964)	equilibrium
FF3	Mkt-RF, SMB, HML	Fama & French (1993)	size and B/M effects (Fama & French (1992))
Carhart	FF3 + MOM	Carhart (1997)	fund evaluation
q-factor	market, size, investment, ROE	Hou, Xue & Zhang (2015)	firms' investment decisions
FF5	Mkt-RF, SMB, HML, RMW, CMA	Fama & French (2015)	dividend-discount model

- FF3 and FF5: the Fama–French three- and five-factor models
  - the FF3 regression is $R^e_{i,t} = \alpha_i + b_i\,\mathrm{MKT}_t + s_i\,\mathrm{SMB}_t + h_i\,\mathrm{HML}_t + \varepsilon_{i,t}$
  - MKT = Mkt-RF; loadings b_i, s_i, h_i
- A good model leaves time-series alphas close to zero, so GRS does not reject it
- FF5 starts from $\frac{M_t}{B_t} = \frac{\sum_{\tau \geq \mathbf{1}} \mathbb{E}[Y_{t+\tau} - dB_{t+\tau}]/(\mathbf{1}+r)^\tau}{B_t}$
  - M: market value, B: book value, Y: earnings, dB: investment
  - at a given M/B, high profitability (RMW) or low investment (CMA) means a high expected return r

Factors beyond FF3

- **RMW** (Robust Minus Weak): high minus low operating profitability
- **CMA** (Conservative Minus Aggressive): low minus high asset growth
- **ROE** (Return On Equity): the profitability factor of the q-factor model
- **MOM** (momentum, also UMD, Up Minus Down)
  - stocks with high minus stocks with low returns over months $t-12$ to $t-2$
  - the last month is skipped
- Past winners keep winning for 3–12 months (Jegadeesh & Titman (1993))
- FF3 is not a special case of FF5
  - the FF3 SMB comes from one sort, the FF5 SMB averages three

Cumulative Returns of the Six Factors, 1963–2026 (1/2)



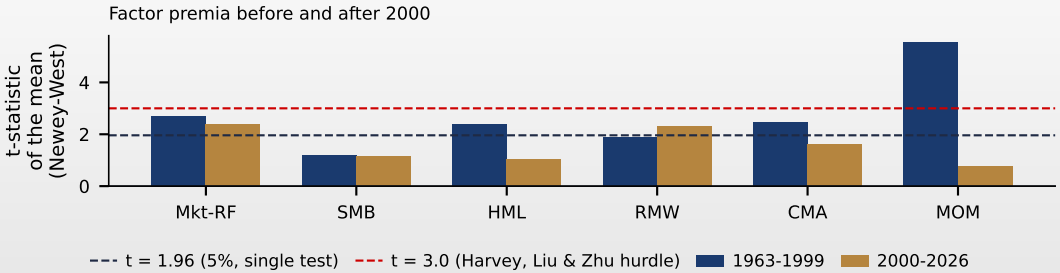
MFM_ch3_factor_cum

Cumulative Returns of the Six Factors, 1963–2026 (2/2)

- Monthly factor returns, compounded from 1, on a log scale
- The factors are excess or long–short returns
  - the curves include no interest on capital or on the collateral of short positions
  - so they are not the wealth of a real investor
- Final value of one unit invested, July 1963 – July 2026
  - Mkt-RF 43.2, MOM 48.0, HML 6.9
  - RMW 5.7, CMA 5.5, SMB 2.9
- Momentum peaked before 2009 and lost heavily when the market rebounded sharply in 2009
  - a **momentum crash**
- The value factor was flat or losing from about 2007 to 2020
- Average annual returns
  - Mkt-RF 7.2%, MOM 7.3%, HML 3.6%
  - RMW 3.1%, CMA 3.0%, SMB 2.3%

MFM_ch3_factor_cum

Do the Premia Survive? (1/2)



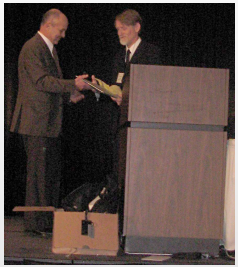
MFM_ch3_factor_tstats

Do the Premia Survive? (2/2)

- Bars: the Newey–West t-statistic of the mean monthly factor return
  - 1963–1999 vs 2000–2026
  - an arbitrary cut, not the publication date
  - the decay after publication: Section 12
- Lines: $t = 1.96$ (5%, a single test) and $t = 3$
  - $t = 3$: the multiple-testing hurdle of Harvey, Liu & Zhu (2016) (Section 8)
- The second period has 319 months against 438
  - so t falls by a factor $\sqrt{319/438} \approx 0.85$ even with an unchanged mean
  - compare the means too
- Momentum: t falls from 5.53 to 0.75
  - the annual mean from 10.7% to 2.5%
- Value (HML): t falls from 2.39 to 1.02
  - the annual mean from 4.2% to 2.8%
- Profitability (RMW) is the exception: t rises from 1.90 to 2.31
- The market premium stays significant in both periods ($t = 2.70$ and 2.39)

MFM_ch3_factor_tstats

The Two Passes



Eugene Fama, Morgan Stanley–AFA (American Finance Association) Award, 2008

Photo: TonyTheTiger (2008); CC BY-SA 3.0; Wikimedia Commons

- The question: are factor exposures rewarded?
  - that is, do betas explain the differences in mean returns, $\mathbb{E}[R_i^e] = \lambda_{\mathbf{0}} + \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$?
- Why Fama & MacBeth (1973): for a traded factor λ is its mean
  - for a non-traded factor (for example, inflation) or for firm characteristics only the cross-sectional regression gives λ
- **Pass 1** (time series, one regression per asset)
  - $R_{i,t}^e = a_i + \boldsymbol{\beta}_i^\top \mathrm{f}_t + \varepsilon_{i,t}$ gives the matrix $\hat{\mathrm{B}}$ $(N \times K)$
- **Pass 2** (cross section, one regression each month on $\mathrm{X} = [1, \hat{\mathrm{B}}]$)

$$R_{i,t}^e = \lambda_{\mathbf{0},t} + \hat{\boldsymbol{\beta}}_i^\top \boldsymbol{\lambda}_t + u_{i,t}, \qquad \hat{\boldsymbol{\lambda}} = \tfrac{1}{T} \textstyle\sum_t \hat{\boldsymbol{\lambda}}_t$$

- $\lambda_{\mathbf{0}}$: zero if the model is right
  - in the Black (1972) version it is the zero-beta portfolio return minus R_f
- a_i: pass-1 intercept; $u_{i,t}$: pass-2 residual
- Why monthly regressions: the variation of the T slopes $\hat{\boldsymbol{\lambda}}_t$ gives the standard error directly (next slide)

Standard Errors for Risk Premia (1/2)

- **Fama–MacBeth**: the monthly $\hat{\lambda}_{k,t}$ form a sample of size T

$$\text{se}_{FM}(\hat{\lambda}_k) = s(\hat{\lambda}_{k,t})/\sqrt{T}$$

  - $s(\cdot)$: the sample standard deviation over the T months
  - robust to correlation of the residuals across assets in the same month
  - if $\hat{\lambda}_{k,t}$ is autocorrelated, use Newey–West on this series
- **Errors in variables**: pass 2 uses estimated betas, not the true ones, so se_{FM} understates the uncertainty

Standard Errors for Risk Premia (2/2)

- The Shanken (1992) correction scales by $(1+c)$ the part of the variance that does not come from factor variability

$$\text{Var}_{\text{Sh}}(\hat{\boldsymbol{\lambda}}) = (1+c)\left[\text{Var}_{FM}(\hat{\boldsymbol{\lambda}}) - \tfrac{1}{T}\boldsymbol{\Sigma}_f\right] + \tfrac{1}{T}\boldsymbol{\Sigma}_f, \qquad c = \hat{\boldsymbol{\lambda}}^\top \boldsymbol{\Sigma}_f^{-1}\hat{\boldsymbol{\lambda}}$$

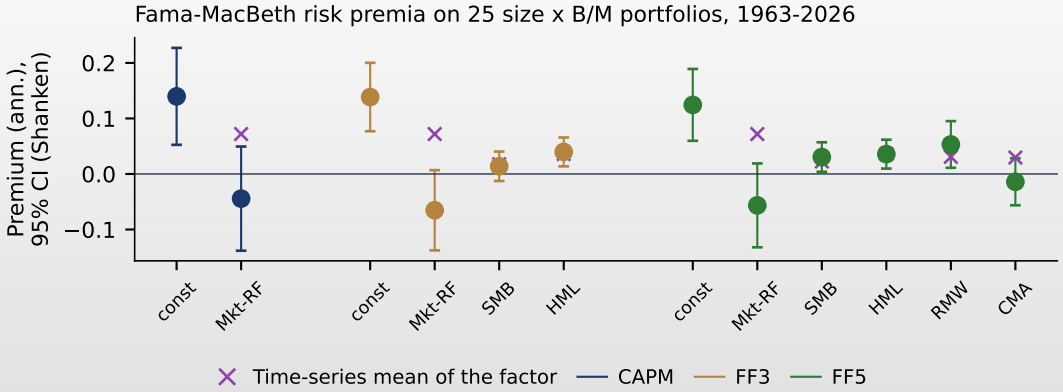
  - Var_{FM}: the Fama–MacBeth variance, $s^2(\hat{\boldsymbol{\lambda}}_t)/T$
  - $\boldsymbol{\Sigma}_f$: factor covariance
  - c: squared Sharpe ratio of the premia, per month (0.03 for FF3)
- Small c (traded factors, monthly data): the correction is small
- Large c (non-traded factors): Fama–MacBeth t-statistics are overstated
- **Misspecified model**: the Fama–MacBeth, Newey–West and Shanken errors assume that the model is correct
  - that is, the pricing errors are zero: $\mathrm{e} = \boldsymbol{\mu} - \mathrm{X}\boldsymbol{\gamma} = 0$
  - $\boldsymbol{\mu}$: mean excess returns, $\boldsymbol{\gamma} = (\lambda_0, \boldsymbol{\lambda}^\top)^\top$
  - if the model is wrong ($\mathrm{e} \neq 0$), the noise in the estimated betas adds variance to the premia
  - the standard errors robust to this are those of Kan–Robotti–Shanken (Kan, Robotti & Shanken (2013); formula: Appendix)

Worked Example: Four Standard Errors for the Value Premium

FF3, HML	$\hat{\lambda}$	SE	t	Significant at 5%?
Fama–MacBeth	3.97%	1.33%	2.99	yes
Newey–West (6 lags)	3.97%	1.62%	2.45	yes
Shanken	3.97%	1.33%	2.99	yes
Kan–Robotti–Shanken	3.97%	1.32%	3.00	yes

- 25 size $\times$ B/M portfolios, 1963–2026
- The Shanken correction leaves the error unchanged, because $c = 0.03$ per month
- What matters most is the autocorrelation of the monthly slopes $\hat{\lambda}_t$
  - Newey–West with 6 lags raises the SE from 1.33% to 1.62%
- For RMW in FF5, Kan–Robotti–Shanken raises the error from 2.1% to 2.6%
  - t falls from 2.5 to 2.1
- The $t > 3$ hurdle (Section 8) is not clearly passed with any standard error

Estimated Risk Premia (1/2)



MFM_ch3_fama_macbeth

Estimated Risk Premia (2/2)

- Bars: 95% confidence intervals with Shanken errors, in annual fractions ($0.1 = 10\%$)
- Crosses: the time-series mean of each traded factor, equal to λ if the model is right
- Estimated premia (25 portfolios, 1963–2026), SE in brackets
  - CAPM: intercept 14.0% (4.4%), market −4.4% (4.8%)
  - FF3: SMB 1.4% (1.4%), HML 4.0% (1.3%)
  - FF5: RMW 5.3% (2.1%), CMA −1.4% (2.2%)
- HML: estimated premium 4.0%, factor mean 3.6%: close
- Market (FF3): estimated premium −6.5%, factor mean 7.2%: far apart
- The cause: market betas differ little across the portfolios
  - 0.86–1.42 in the CAPM and 0.89–1.14 in FF3
- The market premium is therefore **weakly identified**
  - the intercept absorbs the average level of returns (Lewellen, Nagel & Shanken (2010))

MFM_ch3_fama_macbeth

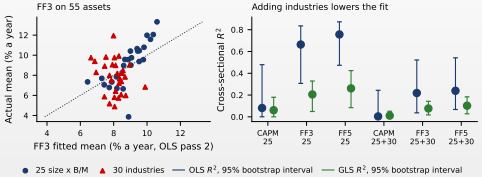
Useless and Weak Factors (1/2) *(deeper)*

- A **useless factor** g_t is independent of all returns
  - its true betas are zero
  - the estimated ones are pure noise, of order $1/\sqrt{T}$
- Pass 2 regresses returns on this noise: $\hat{\lambda}_g$ does not go to zero
  - the t-test too often declares the factor priced when the model is wrong (Kan & Zhang (1999))
- The experiment: on the 25 portfolios, we estimate the CAPM plus an invented factor
  - on these portfolios GRS rejects the CAPM (Section 4)
  - the invented factor is drawn independently from the Normal distribution
  - We repeat the experiment 2 000 times and test at the 5% level whether the invented factor's premium is zero
  - A correct test would reject in about 5% of the repetitions
  - With Fama–MacBeth errors, the false premium looks significant in 60% of the repetitions
  - With Shanken errors, it looks significant in 27% of the repetitions

Useless and Weak Factors (2/2) *(deeper)*

- The remedy: first test in pass 1 $H_0 : \beta_g = 0$ across all N assets
  - $W = \hat{\beta}_g^\top \widehat{\text{Var}}(\hat{\beta}_g)^{-1} \hat{\beta}_g \sim \chi^2_N$
  - if you do not reject, the premium of the factor cannot be estimated
  - The Wald test rejects in only 6.7% of the repetitions, close to 5%
- **Weak identification**: betas close to zero (a weak factor) or nearly equal across assets
  - nearly equal betas: the market, in our example
  - There are tests that are robust to weak identification (Kleibergen (2009))
  - With many assets, consistency of pass 2 depends on factor strength (Anatolyev & Mikusheva (2022))

A Tougher Test: Adding 30 Industry Portfolios (1/2) *(deeper)*



- The 25 portfolios have a strong factor structure (Lewellen, Nagel & Shanken (2010))
  - three factors explain on average 91% of their variation over time (between 79% and 95%)
  - so any factors correlated with SMB and HML reproduce their means
- Left: actual mean (vertical) against the mean predicted by FF3 (horizontal), % a year
  - an exact model would put all points on the diagonal
  - the industries (red triangles) spread vertically

A Tougher Test: Adding 30 Industry Portfolios (2/2) *(deeper)*

- Right panel of the chart: cross-sectional R^2 with 95% bootstrap intervals
  - with 30 industries added, the OLS R^2 of FF3 falls from 0.66 to 0.22
  - and that of FF5 from 0.76 to 0.24
- The FF3 intervals overlap: the fall is suggestive, not precise
  - $[0.31, 0.84]$ on 25 portfolios, $[0.04, 0.52]$ on 55 assets
- The GLS (Generalised Least Squares) R^2 is never above 0.26
  - it weights the errors by the inverse covariance of returns

Three Checks of a Factor Model *(deeper)*

- Three checks (Lewellen, Nagel & Shanken (2010))
  - Broaden the test assets (previous slide)
  - Also report the **GLS** R^2, which weights the errors like GRS
  - Force the premium of each traded factor to equal its mean and drop the intercept: FF3 on the 55 assets gives $R^2 = -0.27$
- Cross-sectional R^2: $R^2 = 1 - \sum_i (\bar{R}_i^e - \hat{\mu}_i)^2 / \sum_i (\bar{R}_i^e - \bar{R}^e)^2$
  - $\hat{\mu}_i$: the predicted mean
  - $\bar{R}^e$: the average of the N assets
- $R^2 < 0$ when the model does worse than the plain average $\bar{R}^e$
  - OLS with an intercept cannot do worse than that
- Individual stocks instead of portfolios (Ang, Liu & Schwarz (2020))
  - betas are more dispersed, but estimated with more noise

Quick Check: Premia, Shanken and a Negative R^2

- A non-traded factor (for example, industrial-production growth), volatility $\sigma_f = 1\%$ a month
- $T = 400$ months; pass 2 gives $\hat{\lambda} = 0.5\%$ a month, with $s(\hat{\lambda}_t) = 3\%$
  1. Compute se_{FM} and the Fama–MacBeth t-statistic
  2. Compute c, the Shanken variance and the Shanken t-statistic. Does the factor clear the $t > 3$ hurdle?
  3. Why would c be tiny for a traded factor with $\sigma_f = 4.5\%$ and the same $\hat{\lambda}$?
- Three portfolios have mean excess returns 2%, 4%, 6% a year
- A model with imposed premia and no intercept fits 6%, 4%, 2%
  4. Compute SST, SSE and R^2
  5. Why can the R^2 of an OLS regression with an intercept never be negative?
- Answers: Appendix

Factors as a Stochastic Discount Factor *(deeper)*



Lars Peter Hansen, Nobel Prize in Economics 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- The question: is a factor needed to price the assets, once we have the other factors?
  - survey: Cochrane (2011)
- **SDF** (Stochastic Discount Factor) m_t: prices all excess returns, $\mathbb{E}[m_t R^e_{i,t}] = 0$
- m_t is high in bad times
  - an asset that pays off exactly then is insurance, so it earns a low mean return
- A linear factor model is an SDF, $m_t = 1 - \mathrm{b}^\top(\mathrm{f}_t - \mathbb{E}\mathrm{f})$
  - b: the SDF coefficients
- Substitute: $\mathbb{E}[R^e_i] = \mathrm{Cov}(R^e_i, \mathrm{f}_t)^\top \mathrm{b} = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$
  - $\boldsymbol{\lambda} = \boldsymbol{\Sigma}_f \mathrm{b}$ (Section 6)
- Two different tests
  - $b_k = 0$: factor k does not help pricing, given the other factors
  - $\lambda_k = 0$: exposure to factor k is not rewarded
  - with correlated factors, $\lambda_k \neq 0$ and $b_k = 0$ can coexist
- b is estimated by GMM (Hansen (1982); Section 4)
  - requiring the sample mean of $m_t(\mathrm{b})\, \mathrm{R}^e_t$ to be zero for every asset

Comparing Models: Maximum Sharpe Ratios (1/2) *(deeper)*

Max. Sharpe ratio (annual)	CAPM	FF3	Carhart	FF5	FF5 + MOM
1963–2026 ($T = 757$)	0.47	0.66	0.97	0.97	1.16
2000–2026 ($T = 319$)	0.48	0.54	0.67	0.97	1.02

- Barillas & Shanken (2018): the better model is the one whose factors allow the highest Sharpe ratio
  - $SR^2(\mathrm{f}) = \boldsymbol{\mu}_f^\top \boldsymbol{\Sigma}_f^{-1} \boldsymbol{\mu}_f$ ($\boldsymbol{\mu}_f$, $\boldsymbol{\Sigma}_f$: means and covariance of the factors)
- The reason: comparing $SR^2(\mathrm{f})$ directly, the test assets no longer matter
  - GRS shows that a model's alphas are small exactly when its factors approach the maximum Sharpe ratio
  - unlike the Hansen–Jagannathan distance, computed on the test assets: ▸ Appendix
- When one model contains the other, in-sample SR^2 can only rise

Comparing Models: Maximum Sharpe Ratios (2/2) *(deeper)*

- Then run a **spanning regression**: regress the added factor on the others
  - a non-zero alpha shows that it brings something new
  - HML on the other five factors: $\alpha = 0.9\%$ a year ($t = 0.75$)
  - since 2000: $\alpha = -1.7\%$ ($t = -0.81$)
  - so no significant incremental contribution of HML, as in Fama & French (2015)
- FF5 vs Carhart (neither contains the other): the difference in annual SR^2 is 0.00
  - 95% interval $[-0.57, 0.66]$
- Since 2000: the difference is 0.50, so again not significant
  - 95% interval $[-0.24, 1.48]$
  - stationary bootstrap, Politis & Romano (1994); Seminar 3, problem B8

The Factor Zoo

- The question: how many of the published factors are real?
  - each was tested as if it were the only one
- Cochrane (2011) called the growing list of factors a *factor zoo*
  - Harvey, Liu & Zhu (2016) catalogue 316 published factors
- Each passed the hurdle $|t| > 1.96$ (5%)
  - if none were real, about $0.05 \times 316 \approx 16$ of 316 tests would pass by chance
- **Data mining** (or p-hacking): searching across many variants until one looks significant
- The Bonferroni correction divides 5% by the number of tests
  - for 316: $p \leq 0.00016$, that is $|t| > 3.78$
- Harvey, Liu & Zhu (2016) propose $|t| > 3.0$ for a new factor
  - the hurdle that follows from the multiple-testing corrections (next slide) applied to all factors tested
- The hurdle rises with every factor published
  - some published factors do not replicate (Section 11)

Controlling False Discoveries (1/2)

- **FDR** (False Discovery Rate): expected share of false rejections among all rejections
- **FWER** (Family-Wise Error Rate): probability of at least one false rejection (Chapter 2)
- Notation of the table
  - M: number of tests; α: the significance level of the tests (here α is not Jensen's alpha)
  - $p_{(\mathbf{1})} \leq \cdots \leq p_{(M)}$: sorted p-values; $H_{(j)}$: the hypothesis with the j-th smallest p-value
  - **BH**: the Benjamini–Hochberg procedure; **BY**: the Benjamini–Yekutieli procedure

Procedure	**Controls**	**Reject $H_{(j)}$ if**	**Valid under**
Bonferroni	FWER	$p_{(j)} \leq \alpha/M$	any dependence
Holm (Holm (1979))	FWER	$p_{(k)} \leq \alpha/(M-k+1)$ for all $k \leq j$	any dependence
BH (Benjamini & Hochberg (1995))	FDR	$j \leq$ largest k with $p_{(k)} \leq k\alpha/M$	positive dependence
BY (Benjamini & Yekutieli (2001))	FDR	as BH, thresholds divided by $\sum_{k=\mathbf{1}}^{M} 1/k$	any dependence

Controlling False Discoveries (2/2)

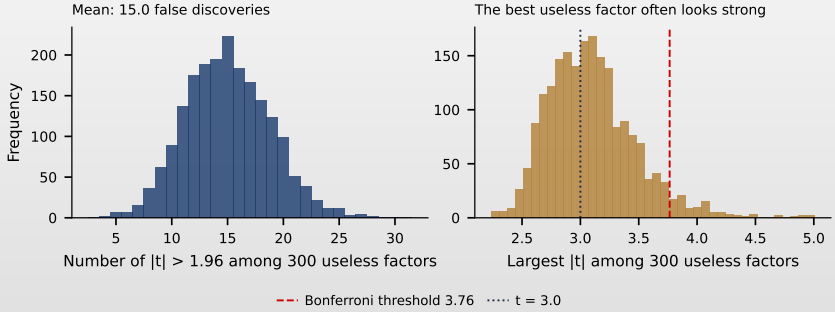
- Positive dependence: the test statistics tend to move up together
  - independence is a special case
  - the technical condition is called PRDS, positive regression dependence on a subset (Benjamini & Yekutieli (2001))
- FWER: suitable when no false discovery is acceptable
- FDR: suitable when searching over many candidate factors
  - a small share of false discoveries is accepted

Worked Example: Ten Candidate Factors

- Ten candidate factors, level $\alpha = 5\%$
  - sorted p-values: 0.0004, 0.002, 0.006, 0.011, 0.018, 0.027, 0.041, 0.070, 0.15, 0.62
- The naive rule rejects every null with $p \leq 0.05$ and finds **7** significant factors
- Bonferroni divides 5% by the 10 tests: only **2** factors pass
  - the threshold is 0.005
- Holm compares the j-th p-value with $0.05/(11 - j)$ and stops at the first one that fails
- The first three p-values pass
  - $0.0004 \leq 0.005$, $0.002 \leq 0.0056$, $0.006 \leq 0.00625$
- The fourth, 0.011, exceeds its threshold 0.0071, so Holm stops at **3** factors
- BH compares the j-th p-value with $0.005\,j$
  - the largest j that passes is 6 ($0.027 \leq 0.030$)
  - so BH declares the first **6** factors significant

Simulation: 300 Useless Factors (1/2)

- The experiment: 300 factors with no true premium
  - each factor's t-statistic is an independent draw from the standard Normal distribution
- We repeat the experiment 2 000 times and count, each time, how many factors pass the hurdle



MFM_ch3_false_discoveries

Simulation: 300 Useless Factors (2/2)

- At the classical hurdle $|t| > 1.96$, on average 15.0 useless factors pass, that is 5% of 300
- At the hurdle $|t| > 3$, on average 0.83 useless factors pass
- In 57% of the repetitions at least one useless factor passes $|t| > 3$
- In half of the repetitions the best useless factor has $|t|$ above 3.06
- The Bonferroni threshold for 300 tests is $|t| > 3.76$
- The right hurdle depends on how many tests were actually tried, unpublished ones included

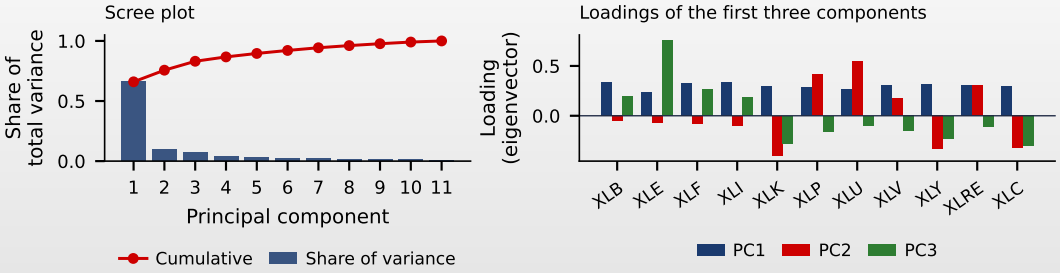
MFM_ch3_false_discoveries

Quick Check: Counting Discoveries

- Five candidate factors, sorted p-values 0.004, 0.012, 0.020, 0.030, 0.200; significance level $\alpha = 5\%$
  1. How many discoveries under the naive rule, Bonferroni, Holm and BH? Write down each threshold you compare with
  2. Under BY the BH thresholds are divided by $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5}$: how many discoveries remain?
  3. If all five nulls were true and the tests independent, what is the probability that at least one p-value falls below 0.05?
  4. Which procedure would you use in a search over many candidate factors?
  5. Which procedure would you use to claim that **no** factor found is spurious?
- Answers: Appendix

PCA of the US Sector ETFs (1/2)

The question: which factors emerge from returns if we do not choose them in advance?



Left (scree plot): each component's share of total variance (bars) and the cumulative share (line); right: the loadings (eigenvectors) of the first three components on each ETF

MFM_ch3_pca

PCA of the US Sector ETFs (2/2)

- **PCA** (Principal Component Analysis)
  - the linear combinations of returns that explain, in turn, the largest part of the remaining variance
  - formula: next slide
- 11 sector ETFs, daily log returns
  - 20 June 2018 – 18 September 2026 ($T = 2073$ common days)
- PC1 explains 66.0% of the variance
  - all loadings positive (0.24–0.34): a near-equal average of the sectors
- PC1 is the market factor: its correlation with SPY is 0.954
- PC2 (9.7%): **defensive versus growth sectors**
  - utilities $+0.55$, staples $+0.42$, real estate $+0.31$
  - vs technology -0.40, consumer discretionary -0.33
- PC3 (7.4%) is dominated by energy
  - the first three components explain 83.1%

MFM_ch3_pca

Principal Components

- **Latent factors**: unobserved factors, estimated from returns alone
  - without saying in advance what they represent
- PCA of the correlation matrix C of standardised returns

$$\mathsf{C} = \mathbf{\Gamma}\mathsf{D}\mathbf{\Gamma}^\top, \qquad \mathsf{y}_t = \mathbf{\Gamma}^\top \mathsf{z}_t$$

  - z_t: standardised returns of day t
  - y_t: the components
  - $\mathbf{\Gamma}$: eigenvectors in columns, the **loadings**
  - D: diagonal matrix of the eigenvalues ℓ_k
- Eigenvalues, sorted: $\ell_1 \geq \cdots \geq \ell_N$ (written ℓ so as not to confuse them with the risk premia λ)
- Component k explains ℓ_k/N of the total variance
- Example, two assets with correlation ρ: $\ell = 1 \pm \rho$
  - the vector $(1,1)/\sqrt{2}$ is the "market"
  - $(1,-1)/\sqrt{2}$ is the difference between the two
- In a crisis ρ rises and the first component dominates

How Many Factors? (1/2)

- Approximate factor model $x_{it} = \mathbf{\Lambda}_i^\top \mathrm{F}_t + e_{it}$
  - both the number of assets N and of periods T are large
  - x_{it}: standardised return; F_t: factors; $\mathbf{\Lambda}_i$: loadings; e_{it}: idiosyncratic part
  - $V(k)$: mean squared residual after the first k components
- Bai & Ng (2002): choose the k that minimises $\ln V(k) + k \frac{N+T}{NT} \ln \min(N, T)$
  - the information criterion IC_{p2}: fit plus a penalty per factor
- Ahn & Horenstein (2013): choose the k where the ratio ℓ_k / ℓ_{k+1} is largest
  - that is, where the eigenvalues drop most sharply
- The rule of Kaiser (1960) ($\ell_k > 1$) has no theoretical justification
  - even pure noise produces eigenvalues scattered around 1

How Many Factors? (2/2)

Panel	N	T	Kaiser	Bai–Ng	Ahn–Horenstein
11 sector ETFs, daily 2018–2026	11	2073	2	5	1
100 size $\times$ B/M portfolios, monthly 1963–2026	95	757	3	4	1

- Standardised returns, with at most $k = 5$ factors for sectors and $k = 8$ for portfolios
  - 5 portfolios with missing months are dropped
- On the 95 portfolios, Ahn–Horenstein picks only the market factor (ratio 14.6)
  - the second peak is at $k = 3$ (ratio 3.5), as in FF3
- Bai–Ng on 11 sectors picks the largest number allowed, $k = 5$
  - its penalty is designed for many assets, so with only 11 the answer is not reliable (Appendix)

Limits of PCA for Asset Pricing

- Loadings are **constant**: PCA cannot follow a stock whose size or value changes
- Components explain **variance**, not necessarily **average returns**
- Individual stocks enter and leave the sample (listings, delistings)
  - so many observations are missing, while classic PCA needs a complete panel
- The sign of each component is arbitrary: flipping an eigenvector changes nothing
- Latent factors are determined only up to a rotation
  - any rotated combination of them fits the data equally well
  - so interpreting them is a choice
- Two extensions of PCA also target average returns
  - risk-premium PCA and the three-pass estimator (Appendix)

Instrumented PCA (1/2) *(deeper)*

- **IPCA**: Instrumented Principal Component Analysis, Kelly, Pruitt & Su (2019)
- Loadings are functions of L observed characteristics $\mathrm{z}_{i,t}$ (size, B/M, momentum, ...)

$$R^e_{i,t+1} = \alpha_{i,t} + \boldsymbol{\beta}^\top_{i,t}\mathrm{f}_{t+1} + \varepsilon_{i,t+1}, \qquad \boldsymbol{\beta}^\top_{i,t} = \mathrm{z}^\top_{i,t}\boldsymbol{\Gamma}_\beta$$

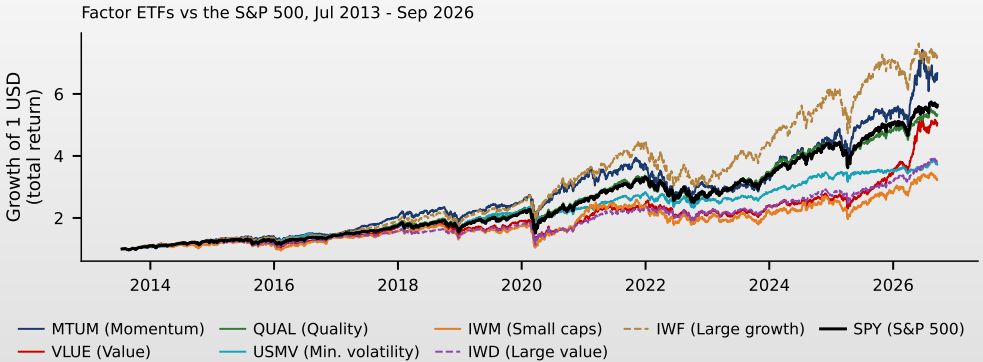
- $\boldsymbol{\Gamma}_\beta$ ($L \times K$) and the factors f_{t+1} minimise $\sum_{i,t}(R^e_{i,t+1} - \mathrm{z}^\top_{i,t}\boldsymbol{\Gamma}_\beta \mathrm{f}_{t+1})^2$
  - by alternating steps
  - with $\boldsymbol{\Gamma}_\beta$ fixed, f_{t+1} comes from a cross-sectional regression
  - with the factors fixed, $\boldsymbol{\Gamma}_\beta$ comes from a panel regression

Instrumented PCA (2/2) *(deeper)*

- The question tested: why do characteristics predict returns?
  - because they measure risk exposure (through beta)?
  - or directly, as alpha ($\alpha_{i,t} = \mathbf{z}_{i,t}^\top \mathbf{\Gamma}_\alpha$)?
- Result on US stocks: a few IPCA factors explain average returns
  - better than models with as many observed factors
- $\mathbf{\Gamma}_\alpha$ is not significant, hence the paper's title: "characteristics are covariances"
- The motivation: IPCA removes two limits of PCA from the earlier slide
  - constant loadings and the need for a complete panel
  - extensions: ▸ Appendix

Factor ETFs versus the Market (1/2)

The question: does an investor who buys the factors through ETFs earn more than with the market?



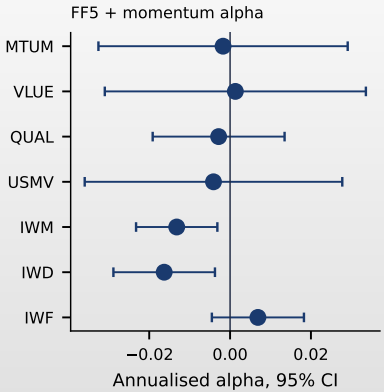
MFM_ch3_factor_etfs

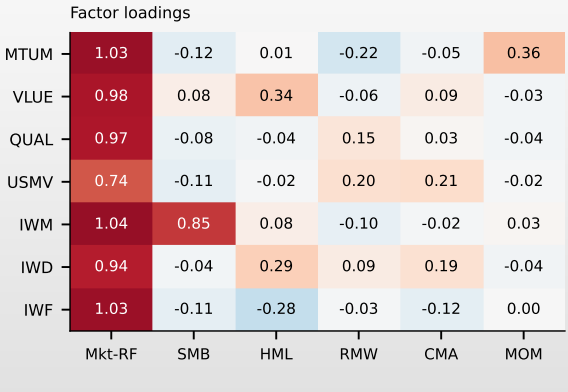
Factor ETFs versus the Market (2/2)

- The chart: the value of 1 USD invested on 18 July 2013
  - dividends reinvested (total return)
  - SPY is the black line
- **Smart beta**: long-only ETFs that buy stocks by fixed rules
  - so they keep market exposure and do not replicate the academic long–short factor
- Russell indices
  - the Russell 1000 holds the 1000 largest US stocks, with value (IWD) and growth (IWF) sub-indices
  - the Russell 2000 holds the next 2000 (IWM)
- **CAGR** (Compound Annual Growth Rate): $(V_T/V_0)^{1/n} - 1$ over n years
- Above SPY (14.0%)
  - large growth (IWF) 16.3%
  - momentum (MTUM) 15.5%
- Below SPY
  - QUAL 13.5%, VLUE 13.1%, IWD 10.7%
  - USMV 10.5%, IWM 9.3%
- Thirteen years are a single historical path
  - too short to reject the factor premia

MFM_ch3_factor_etfs

Alphas of Factor ETFs (1/3)





MFM_ch3_etf_alphas

Alphas of Factor ETFs (2/3)

- The question: do the ETFs earn more than their exposures explain?
  - $R_t^e = \alpha + \beta^\top \mathrm{f}_t + \varepsilon_t$, with the six factors (FF5 + MOM)
- Daily regressions, Newey–West errors
  - each ETF on its own trading days in 2000–2026
  - the chart: annualised alpha with a 95% interval ($0.02 = 2\%$) and the loadings
- The loadings match the labels
  - MTUM on MOM 0.36, VLUE on HML 0.34
  - IWM on SMB 0.85, IWF on HML -0.28
- USMV (minimum volatility) has a market beta of 0.74

MFM_ch3_etf_alphas

Alphas of Factor ETFs (3/3)

- Alphas close to zero
  - MTUM -0.2% ($t = -0.11$)
  - QUAL -0.3% ($t = -0.34$)
- The Russell ETFs have negative alphas
  - IWM -1.3% ($t = -2.57$), IWD -1.6% ($t = -2.54$)
  - fees (below 0.2% a year) explain only a small part
  - the rest comes from the annual Russell reconstitution and from differences with the Fama–French portfolios
- R^2 between 0.81 and 0.97
  - factor ETFs mostly sell exposure to known factors (Seminar 3, problem B11)

MFM_ch3_etf_alphas

Timing, Crowding and Costs

- **Factor timing**: moving capital between factors based on signals
  - for example how cheap a factor is or how it has done recently
  - Premia are uncertain and histories short, so it is hard to show that timing works out of sample (Chapter 13)
- **Crowding**: many investors hold the same factor positions
  - Premia can shrink (McLean & Pontiff (2016))
  - crowding can amplify losses during forced liquidations
  - Momentum crash: after a market fall the momentum portfolio has a negative beta
  - so it loses heavily when the market rebounds sharply, as in 2009 (Daniel & Moskowitz (2016))
- **Implementability**: academic factor returns do not deduct trading costs
  - nor the cost of borrowing the shorted stocks

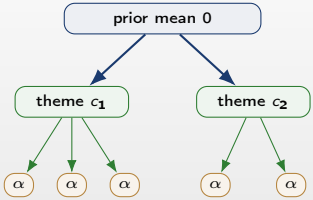
Do Published Factors Replicate?

- **Anomaly**: an average return that the CAPM does not explain
- **Two critiques** of factor research (Jensen, Kelly & Pedersen (2023))
  - Results cannot be reproduced: Hou, Xue & Zhang (2020) replicate only 35% of the anomalies they study
  - Results are the product of testing many factors without correction: Harvey, Liu & Zhu (2016) conclude that most claimed findings are likely false
- **Questions**
  - do published factors have significant CAPM alphas in the original data?
  - are these alphas significant out of sample and in other countries?
  - do they remain significant once multiple testing is taken into account?

Why Replication Matters for This Chapter

- If most factors were false discoveries, multifactor models and factor ETFs (Sections 5 and 10) would rest on noise
- Feng, Giglio & Xiu (2020) show that most new factors add nothing to the existing ones (Appendix)
- Jensen, Kelly & Pedersen (2023) instead see the zoo as many correlated measurements of the same few economic themes
- The quantity tested in the case study is Jensen's alpha (Section 4)
- The classical (non-Bayesian) benchmark is the Benjamini–Yekutieli correction (Section 8)

153 Factors Grouped into 13 Themes *(deeper)*



each factor's alpha borrows strength from the other factors of its theme

- **Data** (Section II of the paper): 153 factors in 93 countries
  - stocks are split into thirds by each characteristic, weighted by market capitalisation, with a cap
  - the factor buys the third with the higher return in the original study and sells the other extreme third
  - each factor is scaled to a specific volatility (uncorrelated with the market) of 10% a year, so that alphas are comparable
- **Themes**: 13 clusters of US factors
  - hierarchical clustering: factors whose CAPM residuals are strongly correlated fall into the same theme
- **Model** (Section I): $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$
  - $f_{i,t}$: return of factor i; r_t^m: market excess return; β_i: market exposure; $\varepsilon_{i,t}$: noise
  - $\alpha_i = c_j + s_n + w_i$: the component of theme j, of signal n (the characteristic, for example B/M) and the component specific to factor i (signal n built in a given country), all with prior mean 0

Shrinking Alphas and the Replication Rate *(deeper)*

- The single-factor case: the estimated alpha is pulled towards zero

$$\mathbb{E}(\alpha \mid \hat{\alpha}) = \kappa \hat{\alpha}, \qquad \kappa = \frac{1}{1 + \sigma^2/(\tau^2 T)}$$

  - equation 5 of the paper; full hierarchy: ▸ Appendix
  - σ: residual volatility
  - τ: dispersion of the true alphas
  - T: number of months
  - short, noisy samples are shrunk more, as in Vasicek (Section 3)
  - σ^2/T plays the role of $\text{se}(\hat{\beta}_i)^2$ and τ^2 that of $\sigma^2_{\hat{\beta}}$

- **Replication rate**: among the factors significant in the original studies
  - the share whose alpha is also significant in the new data
  - in the Bayesian version, a factor is "significant" if its posterior mean divided by its posterior standard deviation is at least 1.96

Most Factors Replicate, but Alphas Fall after the Original Sample (1/3)

Replication rates (Figure 1)

Hou, Xue & Zhang (2020), raw returns	35.0%
Jensen, Kelly & Pedersen (2023), raw returns	55.6%
Raw returns, 119 originally significant	61.3%
CAPM alpha, OLS	82.4%
CAPM alpha, Benjamini–Yekutieli	75.6%
CAPM alpha, empirical Bayes	82.4%
World, empirical Bayes	82.4%

- From 35% (Hou, Xue & Zhang (2020)) to 82.4%
  - With a longer sample, 15 more factors and a different construction, the rate rises to 55.6%
  - Keeping only the factors significant in the original studies, it becomes 61.3%
  - Testing CAPM alphas instead of raw returns, it becomes 82.4%
  - this is the largest step

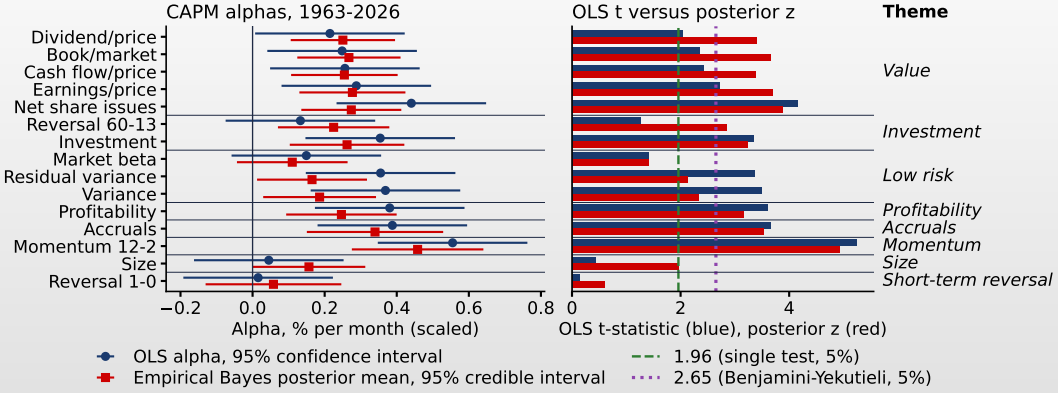
Most Factors Replicate, but Alphas Fall after the Original Sample (2/3)

- Empirical Bayes also gives 82.4%
  - pulling towards zero loses some discoveries
  - information borrowed from the theme adds others, and the effects offset
  - complete results: ▸ Appendix
- The Bayesian FDR among factors with $z \geq 1.96$ is only 0.1%
  - Bayesian FDR: the estimated share of false discoveries
  - although the hurdle stays at 1.96

Most Factors Replicate, but Alphas Fall after the Original Sample (3/3)

- **Decay** after the period of the original study
  - On average a factor's alpha falls from 0.49% a month in-sample (the original study's sample)
  - to 0.26% a month out-of-sample, a 47% drop
  - The slope of out-of-sample alpha on in-sample alpha is 0.26 (Figure 8)
  - an initial alpha higher by 1 percentage point a month stays higher later by only 0.26 percentage points
  - Statistical shrinkage alone would explain a slope of about 0.9, so the real decay is larger than noise
- The strongest factors decay most
  - a sign of arbitrage or of data mining (Section 12)

Do 15 US Factors Replicate? (1/2)



MFM_ch3_jkp_replication

Do 15 US Factors Replicate? Reading the Chart

- Left panel: the CAPM alpha of each factor, in % a month
  - scaled to 10% idiosyncratic volatility a year
- Blue: OLS alpha with its 95% confidence interval
- Red: the empirical-Bayes posterior mean with its 95% credible interval
- Right panel: OLS t (blue) and posterior z (red)
  - z = posterior mean divided by posterior standard deviation
- Dashed lines: 1.96 (a single test, 5%) and 2.65
  - 2.65: the Benjamini–Yekutieli hurdle, 5%
- 15 US long–short factors from value-weighted sorts of the Kenneth French Data Library
  - July 1963 – July 2026 ($T = 757$)
  - grouped by the themes of the paper (authors' cluster labels)

Do 15 US Factors Replicate? (2/2)

- Replication rates
  - OLS: 73.3% (11 of 15 factors)
  - Benjamini–Yekutieli: 53.3% (8 factors; critical $|t| = 2.65$)
  - Empirical Bayes: 86.7% (13 factors)
- Shrinkage pulls alpha towards the theme mean, not only towards zero
- Example: long-term reversal (investment theme) rises from 0.13% to 0.23% a month
  - its OLS t is 1.25, but its posterior z is 2.85
- Not replicated: market beta and short-term reversal ($z \leq 1.41$)
- Size only just passes ($z = 1.96$)
  - in the paper too, size is one of the two themes with a rate below 50%
- The rate here uses all 15 signals
  - without checking whether each was significant in its original study (▸ Appendix)

MFM_ch3_jkp_replication

What the Course Replication Cannot Show

- **Limits of the course version**
  - 15 US signals instead of 153 in 93 countries
  - the portfolios of the Kenneth French Data Library, not thirds built without the smallest firms (micro-caps)
  - a single country, so each signal appears once and the signal component cannot be estimated (its dispersion is set to zero)
  - the result depends on the grouping into themes: if all factors were in one theme, the empirical-Bayes rate would be 100%
- **Limits stated by the authors** (▸ Appendix)
  - the CAPM is the only benchmark
  - all characteristics are built by the same method
  - the prior ignores unpublished factors
- **Open questions** (the paper's data end in 2020)
  - did the alphas and the replication rate hold in 2021–2026?
  - can US factors, used as a prior, improve the estimates of BVB factors, whose histories are short?
- Data and code of the paper: jkpfactors.com

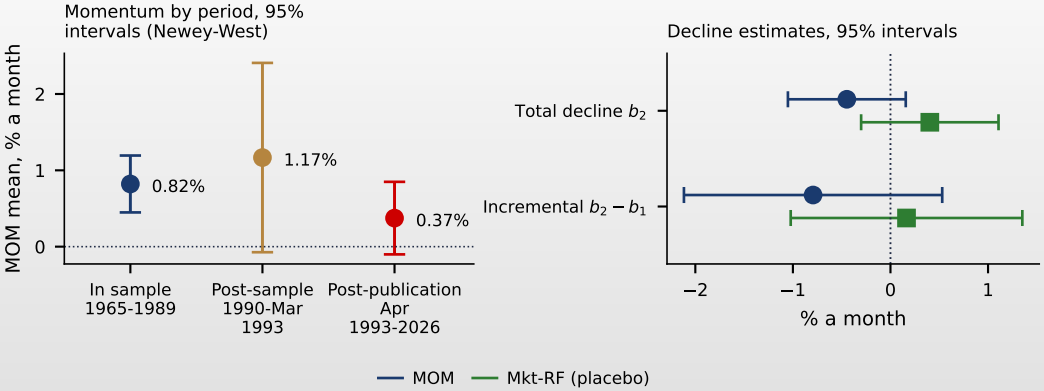
Open Questions and the Discovery Loop

- **Open questions**
  - why do factor premia decay after publication: arbitrage, data mining or a change in risk?
  - does the decay differ across factors and markets?
- Evidence so far
  - McLean & Pontiff (2016): across 97 predictors, returns are 26% lower out of sample and 58% lower after publication
  - the authors attribute the gap to investors trading on the published papers
  - Chen & Zimmermann (2022): open data and code for the published predictors
- **The discovery loop**
  - The AI assistant drafts: a literature table with DOIs (Digital Object Identifiers), competing hypotheses and what each predicts, the regression code, a robustness plan (prompts: ▸ Appendix)
  - The researcher fixes before running: the event dates, the list of factors, the multiple-testing correction and the placebo test (a factor where the effect should not appear)
  - Before interpreting: every DOI resolves in Crossref, the split-sample t-statistics of Section 5 are reproduced, and the version of the French data is recorded

Mini-Case: Was Momentum Arbitraged Away? (1/3)

- Design of McLean & Pontiff (2016): $\text{MOM}_t = a + b_1 \text{PostSample}_t + b_2 \text{PostPub}_t + e_t$
  - monthly, January 1965 – August 2026 ($T = 740$), Newey–West errors
- Dummy variables
  - $\text{PostSample}_t = 1$ in January 1990 – March 1993 (post-sample)
  - $\text{PostPub}_t = 1$ from April 1993 (post-publication)
- a: the in-sample mean (1965–1989)
- b_2: total decline relative to the in-sample mean
- $b_2 - b_1$: incremental effect of publication

Mini-Case: Was Momentum Arbitraged Away? (2/3)



MFM_ch3_ai_discovery

Mini-Case: Was Momentum Arbitraged Away? (3/3)

- In-sample momentum mean: $a = 0.82\%$ a month ($t = 4.32$)
- Decline after publication: $b_2 = -0.45\%$ a month, a mean 54% lower
  - $t = -1.45$, interval $[-1.05\%, 0.16\%]$
- Incremental effect of publication: $b_2 - b_1 = -0.79\%$ ($t = -1.17$)
- Placebo on Mkt-RF: $b_2 = +0.40\%$ ($t = 1.12$)
- The decline is close to the 58% of McLean & Pontiff (2016)
  - but with one factor we cannot reject $H_0 : b_2 = 0$ at 5%
- Arbitrage is one possible interpretation, not a demonstrated cause

Project: Do Factor Premia Fall after Publication?

- **Questions**
  - do the Fama–French factors and momentum decay after publication?
  - does the decay differ across factors?
- **Design**
  - one pooled regression of SMB, HML, RMW, CMA and MOM, with each factor's own dummies for the post-sample and post-publication periods
  - Mkt-RF as placebo
  - errors robust to correlation across factors and to autocorrelation
  - whether the decay is larger where arbitrage is easier can only be tested with stock-level data (size, liquidity, short-sale costs)
- **Replicate first**: the split-sample t-statistics and the mini-case estimates

Project Design: Sample Dates and Deliverables

- **Deliverables**
  - first, an event table: factor, original paper with DOI, end of its sample, publication date, each read from the paper (model row: MOM, Jegadeesh & Titman (1993))
  - a pre-registration note (H_0, H_1, dates from the table, list of factors, corrections), saved in Git (a commit) before seeing the results
  - code as a Quantlet: one folder with the script, its data source and a short description, as in the chapter's Quantlets
  - a report with Newey–West errors, placebo periods and the power of the test
  - power: the probability of rejecting $b_2 = 0$ at 5% if the true decline were, for example, the 58% of McLean & Pontiff (2016)
  - power by simulation or approximately $\Phi(\delta / \mathrm{se}(b_2) - 1.96)$ (δ: the assumed decline; Φ: the distribution function of the standard Normal distribution)
- **Logs**: prompts in `AI_USE.md`, errors caught in `AI_ERRORS.md`

Conclusions: The Steps of a Factor Analysis

1. **Estimate exposures** with HAC errors
   - report sample and proxy
   - shrink extreme betas (Blume (1971), Vasicek (1973))
2. **Test all the alphas at once**
   - GRS measures how much the Sharpe ratio rises when the test assets are added
   - the CAPM fails on the 25 size $\times$ B/M portfolios (flat SML, GRS rejects)
3. **Estimate premia** in two passes
   - with errors that account for estimated betas and for a possibly wrong model
   - a high R^2 on the 25 portfolios proves little
4. **Guard against the zoo**: use hurdles corrected for multiple testing
5. **Check replication**: most published factors replicate
   - but alphas fall by about half after the original sample
6. **Decompose factor ETFs**: they mostly sell exposure to known factors, not alpha

Check Your Understanding

Retrieval questions

- Why does the HAC standard error of a beta differ from the classical one, while the beta does not? (Section 3)
- Why can GRS fail to reject the CAPM on sector ETFs but reject it on the 25 portfolios over the same months? (Section 4)
- Why does a useless factor look priced in pass 2? (Section 6)
- Why can the empirical-Bayes replication rate exceed the Benjamini–Yekutieli rate? (Section 11)

Where the ideas continue

- Chapter 2: a non-zero alpha may be an anomaly or a missing risk factor
  - the joint hypothesis problem
- Chapter 4: factor covariance models and shrinkage for Markowitz
- Chapters 5–6: time-varying volatility, correlation and betas
- Chapter 13: backtest overfitting and machine learning for the cross section of returns

GRS under the Null Hypothesis

- Adding N assets always raises the in-sample Sharpe ratio
  - the test asks whether the rise exceeds what noise would produce
- Sharpe-ratio form: $W = \frac{T-N-1}{N} \frac{\widehat{SR}^2(f,\mathrm{R}) - \widehat{SR}^2(f)}{1+\widehat{SR}^2(f)}$
- Under H_0, with i.i.d. Normal residuals, $W \sim F(N, T-N-1)$, so $\mathbb{E}[W] = \frac{T-N-1}{T-N-3}$
- Multiplying by $\frac{N(1+\widehat{SR}^2(f))}{T-N-1}$: $\mathbb{E}\left[\widehat{SR}^2(f,\mathrm{R}) - \widehat{SR}^2(f)\right] \approx \frac{N\,(1+\widehat{SR}^2(f))}{T-N-3}$
- 25 portfolios, $T = 757$: $25 \times 1.018/729 = 0.035$ (monthly)
  - against $0.405^2 - 0.134^2 = 0.146$ observed
- 9 sector ETFs, $T = 331$: observed gain $0.191^2 - 0.146^2 = 0.015$
  - below the expected $9 \times 1.021/319 = 0.029$

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Tests with Many Assets

- $\hat{\boldsymbol{\Sigma}} = \hat{\mathbf{E}}^\top \hat{\mathbf{E}}/T$ has rank at most $T - 2$ (two parameters per regression)
  - $\hat{\mathbf{E}}$: the $T \times N$ matrix of residuals
  - Example: 100 size $\times$ B/M portfolios on 8 years of monthly data ($T = 96$)
  - then $\hat{\boldsymbol{\Sigma}}$ is singular, GRS does not exist (Seminar 3, problem B7)
- Tests for many assets that avoid $\hat{\boldsymbol{\Sigma}}^{-1}$: standardised sums of squared t-statistics, Pesaran & Yamagata (2024)
- Power enhancement by screening large alphas: Fan, Liao & Yao (2015)

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Misspecification-Robust Standard Errors

- Fama–MacBeth and Shanken errors assume a **correct** model
  - all pricing errors $e = \mu - X\gamma$ are zero ($X = [1, B]$)
- GRS rejects the model here, so γ has no "true" value
  - we estimate the value that minimises the pricing errors (the pseudo-true value)
- The noise in $\hat{B}$ multiplies e and adds a variance term (Kan, Robotti & Shanken (2013))
- Write both passes as one exactly identified GMM system (Section 4)
  - moments $(R_t - a - Bf_t) \otimes (1, f_t)$ and $X^\top(R_t - X\gamma)$
  - the sandwich variance with the full Jacobian is the robust one
  - Setting $e = 0$ in the Jacobian gives the correct-model, heteroskedasticity-robust version of Jagannathan & Wang (1998)
- FF5 on the 25 portfolios (Section 6): the SE of CMA rises from 2.2% to 2.7%
  - that of the intercept from 3.3% to 3.8%
- For HML the change is below 0.05 percentage points
- Advanced course: Generalized Method of Moments

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Hansen–Jagannathan Distance and Bayesian Selection

- **Hansen–Jagannathan distance** (Hansen & Jagannathan (1997))

$$\delta^2 = \min_{\mathrm{b}} \; \mathrm{e(b)}^\top \, \mathbb{E}[\mathrm{R}^e \mathrm{R}^{e\top}]^{-1} \, \mathrm{e(b)}, \qquad \mathrm{e(b)} = \mathbb{E}[m_t(\mathrm{b}) \, \mathrm{R}_t^e]$$

  - how far the model's SDF is from the nearest SDF that prices all assets correctly
  - Excess returns R_t^e with the normalisation $\mathbb{E}[m] = 1$
  - For gross returns R_t: $\mathrm{e} = \mathbb{E}[m\mathrm{R}] - 1$, and the SDF needs a free scale ($\mathbb{E}[m] = 1/(1 + R_f)$)
  - The same weighting matrix for every model, so distances are comparable
  - The test statistic has a weighted χ^2 limit
- Selection among very many models: Bryzgalova, Huang & Julliard (2023)
  - they average over about 2×10^{15} (two quadrillion) factor models
  - robust to weak factors
- Kozak, Nagel & Santosh (2020) shrink the SDF coefficients
  - a few principal components of anomaly returns are enough
- Common lesson: compare all models with the same metric and report its uncertainty

Back

Taming the Zoo with Double-Selection LASSO

- Feng, Giglio & Xiu (2020): does a new factor add information given the existing zoo?
- **Double-selection LASSO** (Least Absolute Shrinkage and Selection Operator)
  - LASSO: a regression whose penalty sets the coefficients of unimportant variables exactly to zero, so it selects variables
  - Step 1: among hundreds of existing factors, LASSO selects those that explain the average returns of the test assets
  - Step 2: a second LASSO selects those correlated with the new factor, so that omitting them does not distort the result
  - Step 3: the SDF coefficient of the new factor is tested, controlling for all factors selected in steps 1 and 2
- Result: most new factors are redundant
- Only a few new factors explain returns that the old ones do not
- The case study reaches a compatible conclusion
  - the zoo is highly redundant, and a few themes carry the information

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Why the Factor-Number Estimators Disagree

- Onatski (2010): a factor is real if its eigenvalue stands clearly outside the range where the noise eigenvalues fall
- For pure noise, the eigenvalues of the correlation matrix fall in $[(1-\sqrt{N/T})^2, (1+\sqrt{N/T})^2]$
  - $= [0.86, 1.15]$ for $N = 11,\ T = 2073$
  - the second eigenvalue, 1.06, lies inside, so the Kaiser rule ($\ell_k > 1$) counts a noise factor
- Bai & Ng (2002): with $N = 11,\ T = 2073$, the IC_{p2} penalty is 0.219 per factor
- On the sectors, the criterion $IC_{p2}(k)$ for $k = 0, \ldots, 5$ is
  - 0, -0.86, -0.97, -1.12, -1.14, -1.17
- It falls all the way to the maximum allowed, $k_{\max} = 5$
  - the penalty is too small to stop weak factors from being added
- Report the range of estimates and the sensitivity to $k_{\max}$, not a single number
  - Seminar 3, problem B3

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Risk-Premium PCA and Three-Pass Premia

- **RP-PCA** (Risk-Premium PCA, Lettau & Pelger (2020)): principal components of $\frac{1}{T}\mathbf{X}^\top\mathbf{X} + \gamma\bar{\mathbf{x}}\bar{\mathbf{x}}^\top$
  - $\mathbf{X}$: the $T \times N$ matrix of excess returns, $\bar{\mathbf{x}}$: their means, γ: the weight on average returns
  - $\gamma = -1$ gives covariance PCA
  - $\gamma > -1$ also favours factors that explain average returns
  - In their data it finds factors with higher Sharpe ratios and smaller pricing errors than PCA
- **Three-pass estimator** (Giglio & Xiu (2021)): the premium of an observed factor g_t when the model omits factors
  - (1) PCA of the test-asset returns
  - (2) cross-sectional regression of the means on the latent betas
  - (3) regression of g_t on the latent factors; the premium of g_t follows from the latent premia and the coefficients of this regression
- Both keep the statistical basis of PCA but aim to explain average returns, not only variance
- IPCA also adds the characteristics of the assets

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IPCA and Crypto-Asset Factors

- IPCA links PCA (statistics) with characteristic sorts (Fama–French): time-varying betas, few factors
- Crypto-assets: market, size and momentum factors (Liu, Tsyvinski & Wu (2022))
  - they price the cross section of cryptocurrencies
- Dimension reduction on the return features of crypto and traditional assets: Pele et al. (2023), Jheng et al. (2026)
- Advanced course: Multivariate Statistical Analysis

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Building the 153 Factors

- **Data** (Section II of the paper)
  - 138 from the list of Hou, Xue & Zhang (2020), plus 15 from the wider literature
  - US returns from 1926, most developed markets from 1986
  - all in US dollars, in excess of the US Treasury bill
- **Construction**, each country and month
  - thirds by the characteristic, with breakpoints computed without the smallest firms (micro-caps)
  - "capped" value weights: market equity winsorised at the NYSE 80th percentile
  - one-month holding period
  - at least 5 stocks per leg and 60 monthly observations
- **Themes**: 13 clusters of US factors (Section II.B)
  - hierarchical clustering, with distance 1− correlation of CAPM residuals and Ward's method (it merges the groups that least increase the variance within groups)

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The Bayesian Hierarchy of Alphas

- **The hierarchy** (Section I of the paper)
  - $\alpha_i = c_j + s_n + w_i$ (theme j, signal n, factor i; eq. 21)
  - prior variance $\mathbf{\Omega} = \mathrm{M}\mathrm{M}^\top \tau_c^2 + \mathrm{Z}\mathrm{Z}^\top \tau_s^2 + \mathrm{I}\tau_w^2$ (eq. 23)
  - posterior mean $(\mathbf{\Omega}^{-1} + T\mathbf{\Sigma}^{-1})^{-1} T\mathbf{\Sigma}^{-1}\hat{\boldsymbol{\alpha}}$ (Proposition 4)
  - M, Z: 0/1 matrices linking each factor to its theme and signal
  - $\tau_c^2, \tau_s^2, \tau_w^2$: variances of the three alpha components
  - $\mathbf{\Sigma}$: residual covariance
  - empirical Bayes: τ_c, τ_s, τ_w by maximum likelihood of $\hat{\boldsymbol{\alpha}} \sim N(0, \mathbf{\Omega} + \hat{\mathbf{\Sigma}}/T)$ (Section I.C.2)

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Complete Results

- With OLS, 98 of the 119 US factors replicate
- The Benjamini–Yekutieli correction requires $|t| > 2.7$ (Figure 4)
- Bayesian FDR: 0.1%, with 95% interval $[0, 1\%]$
- Expected share of true factors: 94% (Section III.C)
- Replication rate for mega and large stocks: 77.3% and 79.8% (Figure 5)
- 11 of 13 themes have a rate of at least 50%
  - the exceptions are low leverage and size
- The high rate among large firms shows that the result does not come only from the smallest firms
- Tangency Sharpe ratio (monthly), US 1972–2020: market 0.13, all factors 0.80 (Figure 14)

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The Course Replication in Detail

- Benjamini–Yekutieli gives the lowest rate, as in the US results of the paper (Figure 1)
- Net share issues are pulled towards the mean of their theme: from 0.44% to 0.27% a month
- The posterior standard deviation is 0.07%–0.10%, below the OLS SE of 0.11%
  - each factor borrows information from the other factors of its theme
- **Prior dispersion**: $\hat{\tau}_c = 0.28\%$, $\hat{\tau}_w = 0.04\%$ a month
  - against 0.35% and 0.21% in Table I of the paper
  - with 15 signals little factor-specific dispersion is left
- **Themes**: the 15 signals fall into 8 of the paper's 13 themes
  - 5 in value, 3 in low risk, 2 in investment

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Limits Stated by the Authors and Extensions

- **Limits stated by the authors**
  - The CAPM is the only benchmark model: low-leverage factors have no significant CAPM alpha, but contribute when combined with the other themes (Section III.A)
  - All characteristics are built by the same rule, sometimes differently from the original paper (Section II.C)
  - Factors tried but never published would shift the prior: in the scenarios of Harvey, Liu & Zhu (2016) the US rate falls to 81.5% and 79.8% (Figure 10)
- **Complete the course replication** (Seminar 3, C3)
  - justify each sign from the original paper
  - compare the dendrogram with the paper's themes
  - add the Bayesian FDR and the post-sample slope (Figure 8)
- Data and code of the paper: jkpfactors.com, github.com/bkelly-lab/ReplicationCrisis

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The Discovery Loop, Reusable Prompts (1/2)

- **Literature**
  - `Summarise the evidence since 2016 on post-publication decay and replication of anomaly returns. For each paper: DOI, sample, number of factors, headline estimate, identification of the decay. Flag entries you are unsure of.`
  - Resolve every DOI on Crossref
  - Check each headline number against the abstract
- **Hypothesis**
  - `Contrast data mining, publication-informed arbitrage and time-varying risk as explanations of decay. For each, describe the predicted pattern across the in-sample, post-sample and post-publication periods, and across high vs low limits-to-arbitrage stocks.`
- **Data and code**
  - `Load the monthly momentum factor from the Kenneth French Data Library; from 1965, regress MOM on a post-sample dummy (1990-01 to 1993-03) and a post-publication dummy (from 1993-04), with Newey-West errors.`
  - Replicate a known number first: momentum $t = 5.53$ before 2000 and 0.75 after (split-sample chart)

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The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**
  - `Write a pre-registered robustness plan: one publication date per factor, a Holm correction across factors, excluding the 2009 momentum crash, the working-paper date instead of the journal date, and 1927-1962 as a placebo period for Mkt-RF, SMB, HML and MOM (RMW and CMA start in July 1963).`
- **Critique**
  - `Referee this: one factor, one publication date, t = -1.45. What is the power to detect a 58% decay? Why does pooling many anomalies change it?`
- **Report**
  - In-sample mean, both dummies with Newey–West errors, the placebo, the data vintage (download date)
  - Pooled factors: errors robust to cross-factor correlation within a month and to autocorrelation
  - Prompts that shaped a decision go into `AI_USE.md`
- The AI assistant proposes the robustness variants
- The researcher chooses in advance which ones are reported

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What the Researcher Must Check

- **Wrong event date**: sample end, working paper and journal issue are different dates
  - test: read them from the paper itself
- **Choosing the factor after seeing it decay**: fix the list of factors in advance
- Apply Holm or BH across all factors
- **Serially correlated, crash-prone returns**: Newey–West errors
- Re-estimate without 2009
- **Data vintages**: the factor library is revised retroactively
  - test: record the download date
  - reproduce the chapter's split-sample t-statistics
- **Low power**: with a single factor, H_0 is rarely rejected
- Report the confidence interval of the decay, not only its sign

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Answers to the Quick Checks (1/2)

Beta, error and shrinkage

- Classical: $0.90 \pm 1.96 \times 0.05 = [0.80, 1.00]$, just below 1
- Newey–West: $0.90 \pm 0.18 = [0.72, 1.08]$, contains 1: no evidence that the stock is less risky
- Report Newey–West: residuals are heteroskedastic and autocorrelated; HAC changes the estimated variance, not the OLS estimate
- $\sigma^2_\beta = 0.10 - 0.02 = 0.08$; SE 0.2: $w = 0.08/0.12 = 0.67$, $\beta^V = 0.67 \times 1.6 + 0.33 \times 1.0 = 1.40$; SE 0.1: $w = 0.08/0.09 = 0.89$, $\beta^V = 1.53$
- The noisier estimate is shrunk more; Blume gives $0.33 + 0.67 \times 1.6 = 1.40$ whatever the precision
- Dimson beta: $0.25 + 0.55 + 0.00 = 0.80$; the stock price absorbs the market return with a one-day delay, so the same-day slope misses the 0.25 part

Reading a GRS statistic

- $W = \frac{60}{4} \times (0.25^2 - 0.15^2)/(1 + 0.15^2) = 15 \times 0.040/1.0225 = 0.59 < 2.53$: not rejected
- Expected gain $4 \times 1.0225/58 = 0.071$, above the observed 0.040: the gap is within noise
- $T = 650$: $W = \frac{645}{4} \times 0.0391 = 6.31 > 2.39$: rejected; the gap is the same, the evidence grows with T

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Answers to the Quick Checks (2/2)

Premia, Shanken and a negative R^2

- $\text{se}_{FM} = 3\%/\sqrt{400} = 0.15\%$, $t = 3.33$
- $c = 0.5^2/1^2 = 0.25$; $\text{Var}_{\text{Sh}} = 1.25 \times (0.0225 - 0.0025) + 0.0025 = 0.0275$, SE 0.166%, $t = 3.02$: only just above 3
- Traded factor: $c = 0.25/4.5^2 = 0.012$, so the correction is negligible
- Mean 4%: SST $= 4 + 0 + 4 = 8$, SSE $= 16 + 0 + 16 = 32$, $R^2 = 1 - 32/8 = -3$; with a free intercept, zero slopes already give SSE = SST, and OLS can only lower SSE

Counting discoveries

- Naive 4; Bonferroni (0.01) 1; Holm: $0.004 \leq 0.01$, $0.012 \leq 0.0125$, $0.020 > 0.0167$ stop: 2; BH thresholds $0.01j$: $0.030 \leq 0.040$ is the last success: 4
- BY: the sum is 2.283, thresholds $0.0044j$: only 0.004 passes: 1
- $1 - 0.95^5 = 0.23$
- A search over candidates: BH (FDR)
- "No factor is spurious": Holm or Bonferroni (FWER)

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Further Study: Courses and Code

Advanced courses on quantinar.com

- Generalized Method of Moments
- Multivariate Statistical Analysis

Code and data

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_03
- Factor data: Kenneth French Data Library

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Acronyms used in this material (1/3)

- **AI**: Artificial Intelligence
- **APT**: Arbitrage Pricing Theory
- **BAB**: Betting Against Beta
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **B/M**: Book-to-Market (ratio)
- **BH**: Benjamini–Hochberg (procedure)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **BY**: Benjamini–Yekutieli (procedure)
- **CAGR**: Compound Annual Growth Rate
- **CAPM**: Capital Asset Pricing Model
- **CMA**: Conservative Minus Aggressive (investment factor)
- **CML**: Capital Market Line
- **DOI**: Digital Object Identifier
- **ETF**: Exchange-Traded Fund
- **FDR**: False Discovery Rate
- **FF3**: Fama–French three-factor model
- **FF5**: Fama–French five-factor model

Acronyms used in this material (2/3)

- **FWER**: Family-Wise Error Rate
- **GLS**: Generalised Least Squares
- **GMM**: Generalized Method of Moments
- **GRS**: Gibbons–Ross–Shanken (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HML**: High Minus Low (value factor)
- **IPCA**: Instrumented Principal Component Analysis
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **MOM**: Momentum (factor)
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **OMV**: Österreichische Mineralölverwaltung — *Austrian Mineral Oil Administration (OMV group)*
- **PC1**: first Principal Component
- **PC2**: second Principal Component
- **PC3**: third Principal Component
- **PCA**: Principal Component Analysis

Acronyms used in this material (3/3)

- **PRDS**: Positive Regression Dependence on a Subset (of p-values)
- **RF**: Risk-Free rate
- **RMSE**: Root Mean Squared Error
- **RMW**: Robust Minus Weak (profitability factor)
- **ROE**: Return On Equity
- **RP-PCA**: Risk-Premium Principal Component Analysis
- **SDF**: Stochastic Discount Factor
- **SE**: Standard Error
- **SMB**: Small Minus Big (size factor)
- **SML**: Security Market Line
- **SSE**: Sum of Squared Errors
- **SST**: Total Sum of Squares
- **UMD**: Up Minus Down (momentum factor)
- **US**: United States